Affine companion groups in characteristic two: scalar realizations and density

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Abstract

We study scalar realizations of affine companion groups in characteristic two. For $q = 2^n$, $d \mid q-1$, and $k = (q-1)/d$, we give an exact Möbius formula and an explicit error for the number of a satisfying $a, 1+a \in C_d$ and $\langle a, 1+a\rangle = C_d$. At the quartic norm boundary $t = 2^s$, $s \geq 2$, and $d = (t+1)(t^2+1)$, the full-family joint-generation count is

$$|\mathcal{R}_d| = \theta_2(d)(t+1)^2 + O(t^{3/2}2^{\omega(d)}), \qquad \theta_2(d) = \prod_{p \mid d}(1-p^{-2}).$$

Each norm fiber has $O(t)$ points. For every nonexceptional fiber we give an explicit $O(t)$ main term with a norm-dependent coefficient and a square-root-scale error; the exceptional fibers are bounded separately. Exact local averaging over all $t+1$ fibers yields the quadratic full-family main term. The number D_d of slopes satisfying $\mathrm{ord}(a) = \mathrm{ord}(1+a) = d$ obeys $D_d \sim (\varphi(d)/d)^2(t+1)^2$, with an explicit error, and is positive for every $s \geq 2$. This exact-order pair is a standard $(t-1)$-primitive-pair object; we do not claim its first formulation. Our contribution is the norm-fiber density analysis, its companion-group interpretation, and the all-dimensional existence proof. We also determine affine generation criteria and prove the optimal uniform index exponent $1/4$; further sections treat spread statistics and a nonlinear example generating A_{16}. Lean checks only the stated triality and field-Singer results; the density and existence arguments use classical curve bounds and exact arithmetic.

1 Introduction

Which primitive affine groups can be generated by a complete mapping and its prescribed companion, and how many slopes realize each group? For a complete mapping f of an elementary abelian 2-group, write $F(f) = (\mathrm{id} \oplus f)^{-1}$. Theorem 5.1 determines the group generated by each affine pair: for $f(x) = Mx \oplus c$, it is $\mathbb{F}_2[M]c \rtimes \langle M, (I+M)^{-1}\rangle$. For scalar maps, this reduces realization to the joint generation condition $\langle a, 1+a\rangle = C_d$ in a specified cyclic subgroup.

The quartic boundary: existence and density. Our principal arithmetic family has $t = 2^s$, $s \geq 2$, $L = \mathbb{F}_{t^4}$, $K = \mathbb{F}_t$, and $d = (t+1)(t^2+1)$. Put $C_d = \ker N_{L/K}$, and let $\mathcal{R}_d$ consist of the slopes $a, 1+a \in C_d$ that jointly generate C_d. Let $\mathcal{D}_d$ be the subset of these slopes for which both orders equal d, and write $D_d = |\mathcal{D}_d|$. Since each full-order element individually generates C_d, we have $\mathcal{D}_d \subseteq \mathcal{R}_d \subseteq \{a \in C_d : 1+a \in C_d\}$. Thus $\mathcal{R}_d$ counts ordinary normalized triangles whose two ratios jointly generate C_d, without requiring either ratio individually to have order d. Theorems 5.23 and 5.24 give

$$|\mathcal{R}_d| = \theta_2(d)(t+1)^2 + O(t^{3/2}2^{\omega(d)}), \qquad D_d \sim \theta(d)^2(t+1)^2,$$

where $\theta_2(d) = \prod_{p|d}(1-p^{-2})$, $\theta(d) = \varphi(d)/d$, and both errors are explicit. Moreover, $D_d > 0$ for every $s \geq 2$. Thus every member of this inner quartic family realizes its primitive affine group with both slopes of full subgroup order.

The count and the all-dimensional existence statement use complementary arguments. The density proof descends to genus-one norm fibers, classifies the constant norm-character exceptions, and proves an $8\sqrt{t}$ mixed-character bound (Proposition 5.21). A fixed fiber has size $O(t)$ and coefficient $\lambda_{t+1}(b)\theta_2(t^2+1)$, which depends on its norm parameter b. Averaging these coefficients gives the quadratic full-family term $\theta_2(d)(t+1)^2$. For existence at every parameter, Proposition 5.25 uses the trace-one construction

$$a = (1+\gamma)^{-1}, \qquad \gamma \in C_{t^2+1}, \qquad \gamma + \gamma^{-1} \in C_{t+1}, \qquad 1 + a = a^{t^2}.$$

The two required orders are tied by Frobenius conjugacy. A separate four-point Lang cover and a two-character sieve prove positivity for every $s \geq 33$; exact field certificates cover $2 \leq s \leq 32$. The intermediate range $33 \leq s \leq 113$ is covered by exact divisor bounds, and a uniform estimate covers every $s \geq 114$. This is a complete parameter partition. The trace-one subfamily itself is empty at $s = 3, 5$; the certificates at these parameters belong to the full family.

Prescribed-order predecessors and quantitative comparison. The exact-order pair above is a $(t-1)$-primitive pair in the standard terminology: $d = (t^4-1)/(t-1)$, and exact order d already implies norm one because $N_{L/K}(a) = a^d$. Thus the norm condition is automatic for D_d. Prescribed-order pairs, their freeness indicators, and their multiplicative-character sieves have established predecessors [12, 13, 15]. The distinction developed here is the effective estimate after descent to the specified norm fibers. For example, [12, Corollary 17], with ambient field size t^4, polynomials $X, X+1$, subgroup indices $t-1, t-1$, and freeness parameters d, d, requires

$$t^2 > (t-1)^2 W(d)^2, \qquad W(d) = 2^{\omega(d)}.$$

This fails for every $t \geq 4$: the coprime odd factors $t+1$ and t^2+1 are both greater than one, so $W(d) \geq 4$. More directly, [15, Proposition 3.1 and Theorem 1] count the same adjacent exact-order pairs: $D_d = N_{t-1}(t^4, 1, 2)$ in their notation. Their sufficient condition at these parameters is

$$t^2 > \frac{((t-1)W(d))^3}{(t-1)W(d) - 1},$$

which also fails for every $t \geq 4$. Our estimates instead exploit the lower-dimensional norm geometry. Failure of these sufficient criteria is not a claim that every possible refinement of the earlier methods has been excluded.

The Frobenius relation in the trace-one construction is classical. The reciprocal-trace reduction in [20, §4] already connects norm-one elements with trace-one elements by a fractional linear change of variable. With its parameter $q = t^2$, [20, Theorem 1.5] gives $\mathrm{ord}(\gamma) = t^2+1$ and $\mathrm{ord}(\gamma+\gamma^{-1}) = t^2-1$. The latter order differs from the required $t+1$; its norm to K has order $t-1$, rather than one. Proposition 5.25 treats this additional reciprocal-trace norm fiber, its four-point conductor, and its prescribed-order count. We do not claim the fractional linear construction or Frobenius coupling as a new mechanism. Hachenberger's quartic projective treatment is a closer geometric predecessor [14, Remark 4.2 and Proposition 7.1]. In characteristic two, $\gcd(t-1, d) = 1$ gives $L^\times = K^\times \times C_d$, so each projective point has a unique representative in C_d; a primitive projective point then corresponds exactly to a representative of order d. Moreover, every primitive ovoid in his partition meets the absolute trace-zero plane in an oval of $t+1$ points. For the construction in

Proposition 5.25, put $x = (1+\gamma)^{-1}$ and $\delta = \gamma + \gamma^{-1}$. Then $x^{t^2} = 1 + x$ and $x^{t^2+1} = \delta^{-1}$. Since $t^2 + 1 \equiv 2 \pmod{t+1}$ and $t+1$ is odd, the C_{t+1}-component of x has order $t+1$; hence $[x]$ lies in a primitive ovoid, and $[x]$ and $[1+x]$ lie in its trace-zero oval. Thus this projective model and ambient oval incidence are prior. The additional condition here is that the paired points both have exact order d and are exchanged by t^2-Frobenius. The cited incidence theorem does not count this arithmetic subfamily. We claim neither the ambient ovoid–plane incidence nor the reciprocal-trace change of variables as new.

The line and translate properties of r-primitive elements were studied in [11]; Remark 4.1 there already allows sufficiently slow growth of r. Varying the index alone therefore does not establish novelty. Choudhary–Sharma [17, Definition 3.1 and Theorem 1] provide a direct predecessor for the exact count D_d. In their notation, set $q = t$, $n = 4$, $m = 2$, $\gamma = 1$, $r_1 = r_2 = t-1$, and $a_1 = a_2 = 1$. Then $r_i' = t-1$, $R_i = d$, $R_i' = 1$, and $Q_{R_i} = d$, so their $N_{1,1}(d)$ is exactly D_d. We therefore do not claim priority for formulating or counting this adjacent exact-order pair. Their Theorem 1 criterion becomes $1 > 2(t-1)^2 W(d)^2$, which fails for every $t \geq 4$. For Theorem 3, write $U = \sum_i u_i$ for the total number of sieved primes and $\Delta = (U-1)/\delta + 2$. When $U = 0$, $\Delta = 1$ and the right-hand side is at least $2(t-1)^2 \geq 18$; when $U \geq 1$, $\Delta \geq 2$ and it is at least $4(t-1)^2 \geq 36$. Theorem 4 also fails for every legal split: its positive denominator is at most 1, its numerator is at least $\sum_i c_i \geq 2$, and the full right-hand side is at least $4(t-1)^2 \geq 36$. Theorem 2 assumes $q \equiv 3 \pmod 4$ and odd n, and the paper's application theorem treats $r = 3$, $n \geq 7$; neither applies here. These are failures of the stated sufficient criteria, not of every possible refinement. The quantitative contribution is the explicit $O(t)$ estimate on individual norm fibers, with their norm-dependent main coefficients, and its exact averaging to the $O(t^2)$ full-family count. The adjacent exact-order statistic is treated as an application, not claimed as a new formulation. The ordinary cyclotomic intersection count is studied, in particular, in [18]; it is not claimed as new. The Lang construction, curve character bound, and primitive indicators are classical [9, 10, 12]. Our proofs verify geometric monodromy, ramification and the resulting errors for the specific realization problem.

Group structure and the other index ranges. The affine group formula gives transitivity, primitivity and double-transitivity criteria (Corollary 5.2), inverse realization and a divisor-sum sieve (Corollaries 5.3 and 5.4). A Jacobi-sum estimate realizes every fixed odd index greater than one in all sufficiently large admissible dimensions (Theorem 5.7), with the exact index-three threshold in Corollary 5.8. Theorem 5.9 and Corollary 5.10 give the uniform range $k \leq q^{1/4}/\sqrt{\log_2 q}$. Norm descent (Theorem 5.11) supplies nonrealizable primitive groups at $k = q^{1/4} + 1$ for every $q = 2^{4s}$ (Corollary 5.12). Thus the uniform exponent $1/4$ is optimal, and the adjacent quartic families $k = q^{1/4} - 1$ and $k = q^{1/4} + 1$ have opposite realization behavior. The universal guarantee still has a logarithmic gap; we do not classify all indices at the boundary. The single full-order count and all-dimensional realization also appear in Theorems 5.19 and 5.20. Primitive inverse classification and generating triangles in generalized Paley graphs are given in Propositions 5.13 and 5.14.

Complete mappings, triality and spread geometry. Complete mappings connect finite groups and Latin-square constructions [1]; over a finite field they also have permutation-polynomial representatives. In characteristic two complete mappings and orthomorphisms coincide. Cycle types and groups generated by their full collections are studied in [25, 26]. Here the individual prescribed pair is the group-generating object. The fixed maps of F correspond to oriented line spreads. An orientation-sensitive Burnside formula (Proposition 7.4) gives, in dimension four, exactly three classes and their stabilizers (Theorem 7.6). The local invariant-orientation criterion in

Proposition 7.5 is a deduction from classical group actions. All six-dimensional standard-spread counts concern one specified underlying spread type.

For the standard Desarguesian spread on $\mathbb{F}_4^m$, semilinear conjugation permutes the line orientations and reverses all their signs. Since the number of lines is odd, the semilinear coset fixes no orientation. This yields a simplified class formula and an explicit asymptotic estimate (Theorem 7.7). The six-dimensional standard family has 80 classes (Corollary 7.8), with all representatives and stabilizer orders supplied in the finite data. The asymptotic argument specializes the classical motion method for distinguishing colorings [30]; existence of regular subsets is studied more generally in [31, 32]. These methods are not claimed as new.

The nonlinear group question is already different from the affine formula: Proposition 8.2 gives a specified pair generating A_{16} whose three companions all have cycle type $1 + 15$. Its proof uses an explicit three-cycle certificate and elementary permutation identities. Proposition 8.3 explains why simply translating a fixed map cannot provide transitive examples beyond dimension two.

Cycle-type existence has also been established in broader settings: Müyesser [29] proves the Friedlander–Gordon–Tannenbaum conjecture for sufficiently large abelian groups. The linked three-map cycle data and the subgroup generated by an individual map and its companion are the specific objects considered here.

The function identity $F^3 = \mathrm{id}$ requires no linearity assumption (Lemma 4.2). In the transversal model of the additive Cayley table, F is induced by the cyclic permutation of the row, column, and symbol coordinates (Proposition 4.3). This places the identity within the classical conjugate, or parastrophic, symmetry of Latin squares [2]. Its fixed maps have a geometric description as oriented line spreads (Proposition 7.2), which includes nonlinear maps. For the entire affine class on $\mathbb{F}_2^n$, Proposition 7.3 gives exact orbit counts. The matrix fixed-space count used there is classical [27, Theorem 2.1]; a short subspace-lattice proof is included to make the application to F self-contained. These statements place the scalar affine classification of Proposition 7.1 in a wider setting.

For the coordinate application, we read the 64 words of $\mathbb{F}_2^6$ through a fixed ordered basis and complement the resulting coordinates. This transformation T is an affine Singer map (layer), meaning that its linear part is a Singer cycle. The associated triple (T, D, K) and the group $\langle T, D\rangle$ then reduce to elementary finite-field statements, with a uniform analogue for $2k$-bit systems (an existence theorem for all k, §6 and Appendix B; the concrete coordinate presentation is verified for $k = 1, 2, 3, 4$). Related families of permutations over $\mathbb{F}_2^n$ with prescribed cycle structure and algebraic degree were studied by Gravel, Panario and Thomson [24]. The underlying even-characteristic existence statement is classical (see [21] and [22, Theorem 1, p. 1]; an equivalent square-order formulation is [20, Theorem 1.5, p. 170]); Cohen and Presern [23] subsequently gave a self-contained account of the prescribed-trace theorem with reduced computation. Appendix B reconstructs the even-characteristic trace-one case and provides an explicit finite certificate; neither the existence theorem nor its character-sum method is claimed as new.

The paper uses this coordinate-permutation symmetry as a common framework for the geometric fixed-map description, the affine orbit counts, and the Singer realization. The geometric classification concerns all complete mappings, whereas the orbit formula concerns the entire affine class, including matrices that are not multiplication in any single chosen field structure. Sections 3–4 give the six-dimensional coordinate data, and Section 5 determines the general affine companion group; Section 6 supplies Singer triples in every even dimension using the classical existence input. Section 7 develops the orbit classifications, including the standard-spread asymptotics. Section 8 studies nonlinear companion groups. Exact finite data and Lean sources with pinned dependencies are supplied with the scope stated in Appendix A. No priority is asserted for the coordinate-permutation symmetry, the classical matrix count, the 56 spreads of $\mathrm{PG}(3, 2)$, or the prescribed-trace theorem. The classical names of the words play no role in any proof.

2 Notation

For the six-bit coordinate example, $V = \mathbb{F}_2^6$; other dimensions are stated explicitly. Coordinates $x = (x_1, \ldots, x_6)$ have x_1 the most significant bit (value 32) and x_6 the least (value 1); a word is identified with its integer value in $\{0, \ldots, 63\}$. Basis $G = (\texttt{63}, \texttt{33}, \texttt{18}, \texttt{49}, \texttt{1}, \texttt{21})$ (a fixed ordered basis — the six words are linearly independent over $\mathbb{F}_2$, by Gaussian elimination; non-uniqueness is discussed in §6); write g_j for the j-th entry, $j = 0, \ldots, 5$. $\mathrm{coords}_G(x) = (c_1, \ldots, c_6)$ is the unique vector with $x = \bigoplus_{i=1}^{6} c_i\, g_{i-1}$, read as the six-bit word $c_1 \cdots c_6$ (c_1 most significant); $\oplus$ is bitwise XOR, i.e. addition in $\mathbb{F}_2^n$. When a field structure is chosen in characteristic two, $\oplus$ is also its field addition. A monic polynomial is called *primitive* if it is irreducible and its roots generate the multiplicative group of its root field. Superscripts on functions and on F denote composition iterates; orders of field elements denote multiplicative orders.

3 T is an affine Singer map

3.1 Definition and basic checks

$T(x) = 63 \oplus \mathrm{coords}_G(x)$. Verified identities: $T(0) = 63$, $T(41) = 0$, $T(38) = 38$; for example $\mathrm{coords}_G(38) = (0, 1, 1, 0, 0, 1)$, read as $011001 = 25$, so $T(38) = 63 \oplus 25 = 38$, and $\mathrm{coords}_G(41) = (1, 1, 1, 1, 1, 1) = 63$, so $T(41) = 63 \oplus 63 = 0$. The integer labels $38, 41, 63$ belong to this chosen coordinate presentation; they are not invariants under changes of basis (see §6).

Lemma 3.1. *T is affine: $T(x) = L_T x + b$ with $b = T(0) = 63$ and $L_T(x) = T(x) \oplus 63$.*

Proof. Since G is a basis, coords_G is an $\mathbb{F}_2$-linear isomorphism. Thus $L_T = \mathrm{coords}_G$, and adding the constant 63 gives the stated affine map. □

Lemma 3.2. $\mathrm{Fix}(T) = \{38\}$*; on the remaining* 63 *points T is a single cycle (type* $1 + 63$*, order* 63*), equivalently L_T has order* 63 *(Proposition 3.3).*

Proposition 3.3. *Identify $(V, +)$ with $\mathbb{F}_{64}$; then L_T is multiplication by a primitive element of $\mathbb{F}_{64}$, a Singer cycle [4]. The characteristic polynomial of L_T is*

$$p_T(t) = t^6 + t^5 + t^3 + t^2 + 1, \tag{1}$$

computed directly as $\det(tI + L_T)$ *(code appendix), primitive of degree* 6*.*

Proof. The matrix in Appendix A gives the displayed determinant. Modulo p_T, repeated squaring and multiplication give

$$t^{64} = t, \qquad \gcd(p_T, t^8 + t) = \gcd(p_T, t^4 + t) = 1.$$

The degree-6 irreducibility criterion, using the prime divisors $2, 3$ of 6, therefore shows that p_T is irreducible. For completeness, $p_T(0) = p_T(1) = 1$ also gives $\gcd(p_T, t^2 + t) = 1$. In $\mathbb{F}_2[t]/(p_T)$ the residue $\bar{t}$ satisfies

$$\bar{t}^{63} = 1, \qquad \bar{t}^{21} = \bar{t}^4 + \bar{t}^2 + \bar{t} + 1 \neq 1, \qquad \bar{t}^{9} = \bar{t}^2 + \bar{t} \neq 1.$$

Since $63 = 3^2 \cdot 7$, its order is 63. The irreducible characteristic polynomial identifies L_T with multiplication by $\bar{t}$ on this field. Finally, $T(38) = 38$ implies $T(x) \oplus 38 = L_T(x \oplus 38)$, proving the fixed-point and cycle claims of Lemma 3.2. □

3.2 Powers and $63 = 7 \cdot 9 = 3 \cdot 21$

Since L_T has order 63, T^d for $d \mid 63$ has type $1 + d \times (63/d)$ (one fixed point plus d cycles of length $63/d$): $T^1 = 1 + 63$, $T^3 = 1 + 3 \cdot 21$, $T^7 = 1 + 7 \cdot 9$, $T^9 = 1 + 9 \cdot 7$, $T^{21} = 1 + 21 \cdot 3$ (verified).

4 Complete mappings and the (T, D, K) triple

A permutation f of an additive group of exponent two is a *complete mapping* if $\mathrm{id} \oplus f$ is again a permutation. Define its *triality operator* $F(f) = (\mathrm{id} \oplus f)^{-1}$, where $(\mathrm{id} \oplus f)(x) = x \oplus f(x)$. By Proposition 3.3, identify the linear part of T with multiplication by a primitive $\alpha \neq 1$. Then $\mathrm{id} \oplus T$ has nonzero slope $1 \oplus \alpha$ and is therefore a permutation. Its inverse $D = F(T)$ has slope $(1 \oplus \alpha)^{-1}$, and $1 \oplus (1 \oplus \alpha)^{-1} = \alpha/(1 \oplus \alpha) \neq 0$, so $\mathrm{id} \oplus D$ is also a permutation. Thus $D = F(T) = (\mathrm{id} \oplus T)^{-1}$ and $K = F(D) = (\mathrm{id} \oplus D)^{-1}$ are defined; since $F^3(T) = T$ below, $\{T, D, K\}$ is the F-orbit of T. Verified facts:

- cycle types $T = D = 1 + 63$, $K = 1 + 7 \cdot 9$ (seven 9-cycles plus one fixed point);
- fixed points $\mathrm{Fix}(T) = \{38\}$, $\mathrm{Fix}(D) = \{41\}$, $\mathrm{Fix}(K) = \{63\}$;
- $F(K) = T$, i.e. $(\mathrm{id} \oplus K)^{-1} = T$; hence $F^3 = \mathrm{id}$ with $T \to D \to K \to T$;
- offset-fixed three-cycle: $D(0) = 38 = \mathrm{Fix}(T)$, $K(0) = 41 = \mathrm{Fix}(D)$, $T(0) = 63 = \mathrm{Fix}(K)$;
- linear parts: for any $n \times n$ matrix A in characteristic two with $p_A(t) = \det(tI + A)$ and $p_A(1) \neq 0$, taking determinants gives

$$p_{(I+A)^{-1}}(t) = \frac{\det\big(t(I + A) + I\big)}{\det(I + A)} = \frac{t^n p_A(1 + t^{-1})}{p_A(1)}.$$

  Here $p_T(1) = 1$ and $p_T(t + 1) = p_T(t)$ (directly, $(t + 1)^6 + (t + 1)^5 + (t + 1)^3 + (t + 1)^2 + 1$ reduces to $t^6 + t^5 + t^3 + t^2 + 1$). Hence $L_D = (I + L_T)^{-1}$ has $p_D(t) = t^6 p_T(1 + t^{-1}) = t^6 p_T(t^{-1}) = t^6 + t^4 + t^3 + t + 1$ (primitive). The reciprocal form uses this shift invariance. L_K has $\Phi_9(t) = t^6 + t^3 + 1$ (cyclotomic, non-primitive, order 9).

Lemma 4.1. *Every complete mapping f of an additive group of exponent two has exactly one fixed point, namely $F(f)(0)$.*

Proof. The equation $f(x) = x$ is equivalent to $x \oplus f(x) = 0$. Since $\mathrm{id} \oplus f$ is bijective, its unique solution is $F(f)(0)$. □

Lemma 4.2. *On complete mappings of any additive group of exponent two, F is defined at every iterate and satisfies*

$$F^2(f) = \mathrm{id} \oplus f^{-1}, \qquad F^3(f) = f.$$

Proof. Put $g = (\mathrm{id} \oplus f)^{-1}$. The identity $x = g(x) \oplus f(g(x))$ gives

$$\mathrm{id} \oplus g = f \circ g.$$

Thus g is complete. Taking inverses yields

$$F^2(f) = (f \circ g)^{-1} = g^{-1} \circ f^{-1} = (\mathrm{id} \oplus f) \circ f^{-1} = \mathrm{id} \oplus f^{-1}.$$

Consequently $\mathrm{id} \oplus F^2(f) = f^{-1}$ and $F^3(f) = f$. Only addition of exponent two and bijectivity were used. □

Proposition 4.3. *For a complete mapping f of an additive group V of exponent two, put*

$$\Gamma_f = \{(x, f(x), x \oplus f(x)) : x \in V\}, \qquad J(f) = f^{-1}.$$

Permuting the three coordinates preserves this family of sets. The permutations $\rho(x, y, z) = (z, x, y)$ and $\sigma(x, y, z) = (y, x, z)$ induce F and J, respectively. Thus J preserves complete mappings and

$$F^3 = J^2 = \mathrm{id}, \qquad JFJ = F^{-1} = F^2.$$

For finite V, Γ_f is a transversal of its additive Cayley table.

Proof. Every triple of Γ_f has sum zero, and each of its three coordinate projections is a bijection to V. Both properties are invariant under coordinate permutations. Conversely these properties recover a unique complete mapping from the first two coordinates. With $z = x \oplus f(x)$, one has $F(f)(z) = x$ and $z \oplus x = f(x)$, so

$$\rho(\Gamma_f) = \Gamma_{F(f)}.$$

With $y = f(x)$, swapping the first two coordinates gives $(y, f^{-1}(y), y \oplus f^{-1}(y))$, hence $\sigma(\Gamma_f) = \Gamma_{J(f)}$. The coordinate permutations satisfy $\rho^3 = \sigma^2 = \mathrm{id}$ and $\sigma\rho\sigma = \rho^{-1}$; injectivity of $f \mapsto \Gamma_f$ transfers these relations to F, J. For finite V, the three bijective projections say exactly that the set contains one cell in each row, each column, and each symbol class of the additive Cayley table. This is the definition of a transversal. □

This is the cyclic part of the classical coordinate-permutation symmetry of Latin squares [2], acting here on transversals of a table invariant under all six coordinate permutations. The proof identifies this relationship explicitly; the order-three symmetry itself is not claimed as new.

For a scalar affine map $f(x) = \alpha x \oplus c$, put $\beta = (1 \oplus \alpha)^{-1}$ and $\kappa = 1 \oplus \alpha^{-1}$. Then $F(f)(x) = \beta x \oplus \beta c$ and $F^2(f)(x) = \kappa x \oplus \alpha^{-1} c$, so the slopes follow $\alpha \mapsto \beta \mapsto \kappa \mapsto \alpha$, with $\alpha\beta\kappa = 1$. The exponent-two hypothesis matters: over $\mathbb{F}_5$, replacing $\oplus$ by ordinary field addition $+$, all three iterates of $f(x) = 2x + 1$ are defined, but $F^3(f)(x) = 2x + 2 \neq f(x)$.

For the Singer example, in field coordinates $T(x) = \alpha x \oplus c$, so $F(T)(0) = D(0) = (1 \oplus \alpha)^{-1} c$ solves $x = \alpha x \oplus c$, i.e. $D(0) = \mathrm{Fix}(T)$, and the slope cycle $\alpha \mapsto (1 \oplus \alpha)^{-1} \mapsto 1 \oplus \alpha^{-1} \mapsto \alpha$ gives the two remaining identities. For the six-bit layer the values are $F(T)(0) = 38$, $F(D)(0) = 41$, $F(K)(0) = 63$. Conceptually, T is the launching map — an affine Singer layer read through the fixed generator basis (a change of basis) — while the triality operator F keeps the third-order cycle running: $T \mapsto D \mapsto K \mapsto T$, carrying the fixed point $38 \mapsto 41 \mapsto 63 \mapsto 38$.

5 Groups generated by affine companions

For a vector u write $\tau_u(x) = x \oplus u$. If $U \leq V$ is invariant under $M \leq \mathrm{GL}(V)$, we realize $U \rtimes M$ as $\{x \mapsto Lx \oplus u : L \in M,\ u \in U\}$ in the given coordinates. Composition has the usual rule $(u, L)(v, N) = (u \oplus Lv, LN)$.

Theorem 5.1 (Affine companion group)**.** *Let $V = \mathbb{F}_2^n$, $n \geq 1$, and let $f(x) = Ax \oplus c$ be complete, so $A, I + A \in \mathrm{GL}(n, 2)$. Put*

$$B = (I + A)^{-1}, \qquad R = \mathbb{F}_2[A], \qquad M_A = \langle A, B \rangle,$$
$$U = Rc = \mathrm{span}_{\mathbb{F}_2}\{c, Ac, \ldots, A^{n-1}c\}.$$

For every $c \in V$, including $c = 0$,

$$\langle f, F(f) \rangle = U \rtimes M_A, \qquad |\langle f, F(f) \rangle| = 2^{\dim U} |M_A|.$$

In particular its pure translations are exactly $\{\tau_u : u \in U\}$.

Proof. The minimal polynomial of A is coprime to t and $t+1$. Bezout identities therefore show that A^{-1} and B belong to R; their inverses do as well. Thus $M_A \le R^\times$ is abelian and preserves U. Cayley–Hamilton gives the displayed finite spanning set for U. Since $g = F(f)$ has the explicit form $g(x) = Bx \oplus Bc$, both generators lie in $U \rtimes M_A$, proving one inclusion.

Using $AB = I + B$ and $[f, g] = f \circ g \circ f^{-1} \circ g^{-1}$, direct composition gives

$$g \circ f(x) = ABx, \qquad f \circ g(x) = ABx \oplus Bc, \qquad [f, g] = \tau_{Bc}.$$

Conjugation by f sends τ_u to τ_{Au}. Hence the generated group contains all translations whose vectors lie in $\operatorname{span}\{A^j Bc : j \ge 0\} = RBc = Rc = U$, where B is a unit of R. It contains $\tau_c \circ f = A$ and $\tau_{Bc} \circ g = B$ as pure linear maps, so it contains M_A as well. This proves the reverse inclusion. Distinct pairs (u, L) give distinct affine maps, yielding the order and translation statements. □

Corollary 5.2 (Action criteria)**.** *In Theorem 5.1, write $H = \langle f, F(f)\rangle$. Then H is transitive on V if and only if $U = V$, equivalently c is a cyclic vector for A. It is primitive if and only if $c \ne 0$ and the characteristic polynomial of A is irreducible. If A is identified with multiplication by $\alpha \in \mathbb{F}_{2^n}$ in this irreducible case, then for $c \ne 0$*

$$|H| = 2^n \operatorname{lcm}\big(\operatorname{ord}(\alpha), \operatorname{ord}(1 \oplus \alpha)\big).$$

It is doubly transitive if and only if $c \ne 0$, A is irreducible, and this least common multiple is $2^n - 1$. In that case $H = \mathrm{AGL}(1, 2^n)$ in these field coordinates.

Proof. The orbit of 0 is exactly U, giving transitivity. When $U = V$, the group contains the full regular translation group and its stabilizer of 0 is M_A. A block containing 0 is an additive subspace: if u belongs to the block, its translate by u meets the block at u, so the block property makes that translate equal to the block. The block is also M_A-invariant. Conversely the cosets of any M_A-invariant subspace form a block system (cf. [19]). An A-invariant subspace is invariant under the polynomial maps B, A^{-1}, B^{-1}, so the M_A-invariant subspaces are exactly the A-invariant ones. The action is therefore primitive precisely when A acts irreducibly; this is equivalent to an irreducible characteristic polynomial. Such an A makes every nonzero vector cyclic, which explains the condition $c \ne 0$.

An irreducible A identifies V with $\mathbb{F}_{2^n}$ and A with multiplication by α; the two multipliers generate $\langle \alpha, (1 \oplus \alpha)^{-1}\rangle$ in the cyclic multiplicative group. Its order is the least common multiple of the two element orders. Double transitivity means that M_A is transitive on $V \setminus \{0\}$, which holds exactly when this subgroup is the entire multiplicative group. Double transitivity implies primitivity, so the conditions are also necessary outside the irreducible case. □

Corollary 5.3 (Realizing prescribed scalar affine subgroups)**.** *Fix $n \ge 2$, put $q = 2^n$, and regard $\mathbb{F}_q$ as an n-dimensional vector space over $\mathbb{F}_2$. For a divisor $d > 1$ of $q - 1$, let C_d be the unique subgroup of order d of $\mathbb{F}_q^\times$, and define*

$$\mathcal{R}_d = \{a \in C_d : \ 1 + a \in C_d, \ \langle a, 1 + a\rangle = C_d\}.$$

The particular affine subgroup $H_d = \mathbb{F}_q^+ \rtimes C_d \le \mathrm{AGL}(1, q)$ is generated by f and $F(f)$ for a scalar affine complete mapping $f(x) = ax + c$ if and only if

$$\operatorname{ord}_d(2) = n \quad \text{and} \quad \mathcal{R}_d \ne \varnothing.$$

Here $\operatorname{ord}_d(2)$ denotes the multiplicative order of 2 modulo the odd integer d. When the conditions hold, the exact number of scalar affine maps f realizing this specified subgroup H_d (in these fixed field coordinates) is

$$(q - 1)|\mathcal{R}_d|.$$

Proof. If $\langle f, F(f)\rangle = H_d$, Theorem 5.1 shows that its linear part is $\langle a, (1+a)^{-1}\rangle = C_d$, so $a \in \mathcal{R}_d$. Its pure translations fill the field, which forces $c \neq 0$ and $\mathbb{F}_2[a] = \mathbb{F}_q$. Since $C_d \leq \mathbb{F}_q^\times$ and $\langle a, 1+a\rangle = C_d$, the least subfield containing C_d is $\mathbb{F}_2[a]$. The subfield generated by a multiplicative subgroup of order d has degree $\mathrm{ord}_d(2)$ over $\mathbb{F}_2$; thus this order is n.

Conversely, suppose $\mathrm{ord}_d(2) = n$ and $a \in \mathcal{R}_d$. Because $1 + a \in \mathbb{F}_2[a]$ and $C_d = \langle a, 1 + a\rangle$, the field $\mathbb{F}_2[a]$ contains C_d, hence equals $\mathbb{F}_q$. In particular $a \neq 0, 1$, multiplication by a is irreducible over $\mathbb{F}_2$, and for any $c \neq 0$ one has $\mathbb{F}_2[a]c = \mathbb{F}_q$. Theorem 5.1 now gives $\langle f, F(f)\rangle = H_d$. There are exactly $|\mathcal{R}_d|$ admissible slopes and $q-1$ admissible translations; different pairs (a, c) define different maps. □

Scope. This is an exact realizability test for specified scalar affine subgroups, not a classification of all subgroups of $\mathrm{AGL}(n, 2)$ up to conjugacy. The remaining arithmetic question is to determine the nonemptiness and size of $\mathcal{R}_d$ uniformly in n, d; the criterion by itself is a consequence of Theorem 5.1.

Corollary 5.4 (An arithmetic sieve for realizable slopes)**.** *Retain the notation of Corollary 5.3, and for every divisor e of $q - 1$ set*

$$T_e = |\{a \in C_e : 1 + a \in C_e\}|.$$

Then the number of slopes realizing C_d satisfies

$$|\mathcal{R}_d| = \sum_{e|d} \mu(d/e) T_e, \tag{2}$$

where μ is the number-theoretic Moebius function. In fact,

$$T_e = \deg \gcd_{\mathbb{F}_2[t]}\big(t^e + 1, (t + 1)^e + 1\big), \tag{3}$$

so these numbers depend on e, not on the ambient field containing C_e. Moreover, for $d > 3$ one has

$$|\mathcal{R}_d| \equiv 0 \pmod 3;$$

for $d = 3$ (necessarily n even), $|\mathcal{R}_3| = 2$. In particular, the full-translation subgroup $\mathbb{F}_{2^n}^+ \rtimes C_3$ is realizable by a scalar affine companion pair in fixed field coordinates exactly when $n = 2$.

Proof. Every pair $(a, 1 + a)$ counted by T_e generates a unique subgroup $C_s \leq C_e$. Therefore $T_e = \sum_{s|e} |\mathcal{R}_s|$, extending the definition by $\mathcal{R}_1 = \varnothing$. Moebius inversion yields (2). Since $e \mid q - 1$, both polynomials in (3) split into simple linear factors over $\mathbb{F}_q$ (their derivatives are nonzero, as e is odd). Their common roots are precisely the elements a counted by T_e, proving the polynomial-gcd formula.

The fractional-linear transformation $\varphi(a) = (1+a)^{-1}$ preserves $\mathcal{R}_d$: indeed $\varphi(a)$ and $1+\varphi(a) = a/(1 + a)$ belong to C_d and generate exactly the subgroup generated by a and $1 + a$. Direct simplification gives $\varphi^3(a) = a$. A fixed point obeys $a^2 + a + 1 = 0$, hence has multiplicative order three and can lie in $\mathcal{R}_d$ only when $d = 3$. Thus for $d > 3$ all φ-orbits in $\mathcal{R}_d$ have length three. For $d = 3$, the two nonidentity elements of $\mathbb{F}_4^\times$ comprise $\mathcal{R}_3$. Finally $\mathrm{ord}_3(2) = 2$, so Corollary 5.3 gives the last claim. □

As a symbolic nonrealizability example in dimension six,

$$\gcd(t^9 + 1, (t + 1)^9 + 1) = t^2 + t + 1,$$

while the analogous gcd for exponent 21 is

$$(t^2+t+1)(t^3+t+1)(t^3+t^2+1).$$

Together with the divisor-sum formula, these identities give $|\mathcal{R}_9| = 2-2 = 0$ and $|\mathcal{R}_{21}| = 8-6-2 = 0$. Although $\mathrm{ord}_9(2) = \mathrm{ord}_{21}(2) = 6$, neither $\mathbb{F}_{64}^+ \rtimes C_9$ nor $\mathbb{F}_{64}^+ \rtimes C_{21}$ can be realized by a scalar affine companion pair. These are exact polynomial identities, not claims of a complete nonrealizability classification.

The quantities T_e are standard cyclotomic intersection counts. For example, if $k = (q-1)/e$ and χ is a multiplicative character of order k on $\mathbb{F}_q$, extending all powers (including the trivial character) by zero at zero, then

$$T_e = \frac{1}{k^2} \sum_{i,j=0}^{k-1} \sum_{x \in \mathbb{F}_q} \chi^i(x)\chi^j(1+x).$$

This identity is simply the multiplicative-character indicator of C_e. Accounting separately for the trivial-character terms gives the effective classical estimate in Proposition 5.6; its combination with a maximal-subgroup sieve will prove that $\mathcal{R}_d$ is nonempty in the families below.

Theorem 5.5 (An infinite family of primitive nonrealizable groups)**.** *Let $r \geq 2$, $q = 2^{2r}$, and $d = 2^r + 1$. The scalar affine group*

$$H_d = \mathbb{F}_q^+ \rtimes C_d \leq \mathrm{AGL}(1, q)$$

is primitive of degree q, but no affine complete mapping f on $\mathbb{F}_2^{2r}$ satisfies $\langle f, F(f)\rangle = H_d$. The same nonrealizability holds for every $\mathrm{GL}(2r, 2)$-conjugate of this subgroup. More generally, if an odd integer $d > 3$ divides $2^r + 1$, then $\mathcal{R}_d = \varnothing$ in every characteristic-two field containing C_d.

Proof. Since $2^r \equiv -1 \pmod{d}$, the order of 2 modulo d divides $2r$. It cannot be at most r, as $0 < 2^j - 1 < d$ for $1 \leq j \leq r$. Every proper divisor of $2r$ is at most r, so $\mathrm{ord}_d(2) = 2r$. Consequently the multiplicative action of C_d on $\mathbb{F}_q$ is irreducible over $\mathbb{F}_2$, and its affine extension by all translations is primitive.

More generally, suppose $d > 3$ divides 2^r+1 and $a, 1+a \in C_d$. Because $a^d = 1$ and $(1+a)^d = 1$, we have $a^{2^r} = a^{-1}$ and

$$1 = (1+a)^{2^r+1} = (1+a^{-1})(1+a) = a + a^{-1}.$$

Multiplying by a yields $a^2+a+1 = 0$. Thus a and $1+a = a^2$ both have order 3, and $\langle a, 1+a\rangle = C_3$. Since $d > 3$, no a belongs to $\mathcal{R}_d$. Corollary 5.3 proves that no scalar affine companion pair generates H_d. Moreover, every affine permutation belonging to the specified group H_d has precisely the scalar form $x \mapsto ax+c$, so this excludes arbitrary affine complete mappings whose generated group equals H_d.

Finally, for an invertible $\mathbb{F}_2$-linear change of coordinates S, one has $F(SfS^{-1}) = SF(f)S^{-1}$. Hence an affine companion realization of a conjugate subgroup would conjugate back to one of H_d, which is impossible. □

The theorem gives an obstruction beyond irreducibility and primitivity; it does not claim that all nonrealizable primitive affine groups have a complement of this form.

Proposition 5.6 (A positive lower bound for realizing slopes)**.** *Let $q = 2^n$, let $d > 1$ divide $q - 1$, and set $k = (q - 1)/d$. Write $\omega(d)$ for the number of distinct prime divisors of d, and put*

$$b_k = \frac{(k-1)(k-2)}{k^2}.$$

For every divisor e of $q - 1$, with $\ell = (q - 1)/e$, one has

$$\left| T_e - \frac{q + 1 - 3\ell}{\ell^2} \right| \leq b_\ell \sqrt{q}. \tag{4}$$

Consequently,

$$|\mathcal{R}_d| \geq \frac{q}{k^2} \left(1 - \sum_{p \mid d} \frac{1}{p^2} \right) - \frac{3k - 1}{k^2} - (\omega(d) + b_k)\sqrt{q}, \tag{5}$$

where the sum runs over the distinct prime divisors of d. In particular,

$$|\mathcal{R}_d| > \frac{3q}{4k^2} - \frac{3k - 1}{k^2} - (\omega(d) + b_k)\sqrt{q}. \tag{6}$$

Proof. The input is the classical Jacobi-sum calculation [5, Theorems 1–2]; we include its character-sum proof. Choose a character χ of order ℓ, and extend all its powers, including the trivial character, by zero at zero. The subgroup indicator gives

$$T_e = \frac{1}{\ell^2} \sum_{i,j=0}^{\ell-1} J(\chi^i, \chi^j), \qquad J(\eta, \theta) = \sum_{x \in \mathbb{F}_q} \eta(x)\theta(1 + x).$$

The term $i = j = 0$ is $q - 2$. Each of the $2(\ell - 1)$ terms with exactly one nontrivial character is -1. The $\ell - 1$ terms with both characters nontrivial and product trivial are also -1: the substitution $u = x/(1 + x)$ identifies their sum with the sum of a nontrivial character over $\mathbb{F}_q^\times \setminus \{1\}$.

For completeness, fix a nontrivial additive character ψ and set $G(\eta) = \sum_{x \neq 0} \eta(x)\psi(x)$. For nontrivial η, substituting $x = ty$ into its squared modulus gives

$$|G(\eta)|^2 = \sum_{t \neq 0} \eta(t) \sum_{y \neq 0} \psi((t + 1)y) = (q - 1) - \sum_{t \neq 0,1} \eta(t) = q.$$

If η, θ, and $\eta\theta$ are all nontrivial, grouping $G(\eta)G(\theta)$ by $z = x + y$ gives

$$G(\eta)G(\theta) = J(\eta, \theta)G(\eta\theta).$$

Here the $z = 0$ contribution vanishes by multiplicative-character orthogonality; for $z \neq 0$ put $x = zu$, $y = z(1 + u)$. Thus $|J(\eta, \theta)| = \sqrt{q}$. There are $(\ell - 1)(\ell - 2)$ remaining terms. Their triangle bound, together with the terms evaluated above, proves (4), also when $\ell = 1$.

A pair counted by T_d that does not generate C_d lies in at least one maximal subgroup $C_{d/p}$, for a prime $p \mid d$. Therefore

$$|\mathcal{R}_d| \geq T_d - \sum_{p \mid d} T_{d/p}.$$

Applying (4) to the first term gives $T_d \geq (q + 1 - 3k)/k^2 - b_k\sqrt{q}$. For the subgroup of index kp, the same bound gives

$$T_{d/p} \leq \frac{q + 1 - 3kp}{k^2 p^2} + b_{kp}\sqrt{q} \leq \frac{q}{k^2 p^2} + \sqrt{q}.$$

This proves (5). All the primes dividing d are odd. The elementary telescoping bound

$$\sum_{p|d} \frac{1}{p^2} \le \sum_{j=1}^{\infty} \frac{1}{(2j+1)^2} < \frac{1}{4} \sum_{j=1}^{\infty} \frac{1}{j(j+1)} = \frac{1}{4}$$

proves (6). □

Theorem 5.7 (Realization at any fixed odd index)**.** *Let $k > 1$ be odd. For every sufficiently large n satisfying $k \mid 2^n - 1$, put $q = 2^n$ and $d = (q-1)/k$. Then*

$$H_d = \mathbb{F}_q^+ \rtimes C_d$$

is a primitive subgroup of index k in $\mathrm{AGL}(1, q)$ and is generated by f and $F(f)$ for a scalar affine complete mapping. An explicit sufficient condition is $d > \sqrt{q} - 1$ and

$$L_{q,k,d} := \frac{3q}{4k^2} - \frac{3k-1}{k^2} - (\omega(d) + b_k)\sqrt{q} > 0. \tag{7}$$

Under these conditions the number of realizing maps in the specified field coordinates is $(q-1)|\mathcal{R}_d| > (q-1)L_{q,k,d}$. In particular, each fixed odd index $k > 1$ gives an infinite family of realizable proper primitive affine groups.

Proof. The degree $s = \mathrm{ord}_d(2)$ of the field generated by C_d divides n. If $s < n$, then $s \le n/2$ and $d \mid 2^s - 1 \le \sqrt{q} - 1$, contrary to the assumed inequality. Hence $s = n$. Proposition 5.6 and (7) imply $\mathcal{R}_d \neq \varnothing$. Corollaries 5.3 and 5.2 give the realization, primitivity, and exact count. The index is k because the translation subgroup is full and the complement has order $(q-1)/k$.

To establish the infinite-family assertion without an arithmetic assumption on d, note that $3^{\omega(d)} \le d < 2^n$. Since $3 > 2^{3/2}$, this yields $\omega(d) < 2n/3$. Also $0 \le b_k \le 1$. For fixed k, the positive term $3 \cdot 2^n/(4k^2)$ in (7) eventually exceeds the constant term and $(2n/3+1)2^{n/2}$. Likewise $(2^n-1)/k > 2^{n/2}-1$ eventually. There are infinitely many admissible n, namely multiples of $\mathrm{ord}_k(2)$. The sufficient conditions therefore hold for every sufficiently large admissible dimension. □

Corollary 5.8 (The exact threshold at index three)**.** *For even $n \ge 4$, the specified subgroup*

$$\mathbb{F}_{2^n}^+ \rtimes C_{(2^n-1)/3}$$

is primitive and is generated by a scalar affine complete mapping and its prescribed companion if and only if $n \ge 8$. Thus index-three proper primitive companion groups exist in every even dimension from eight onwards.

Proof. For every even $n \ge 4$, writing $t = 2^{n/2} > 2$ gives $(t^2-1)/3 > t-1$. The field-degree argument in Theorem 5.7 proves primitivity. At $n = 4$ the complement has order $5 = 2^2 + 1$, excluded by Theorem 5.5. At $n = 6$ it has order 21, and the polynomial identities following Corollary 5.4 give $\mathcal{R}_{21} = \varnothing$.

For $k = 3$, $b_3 = 2/9$. Using $\omega(d) < 2n/3$, the uniform lower bound in (6) is greater than

$$\frac{2^n}{12} - \frac{8}{9} - \left(\frac{2n}{3} + \frac{2}{9}\right) 2^{n/2}.$$

This expression equals $424/3 > 0$ at $n = 14$. After division by $2^{n/2}$, increasing n by two doubles its positive exponential term, whereas its linear term increases by $4/3$ and its negative exponential

term decreases in magnitude. Its increase is greater than $2^{n/2}/12 - 4/3 > 0$ for even $n \geq 14$, proving positivity there. For $n = 10$, $d = 341 = 11 \cdot 31$ and $\omega(d) = 2$; for $n = 12$, $d = 1365 = 3 \cdot 5 \cdot 7 \cdot 13$ and $\omega(d) = 4$. The bound $L_{q,3,d}$ equals $40/3$ and $632/9$, respectively, so these dimensions are covered as well.

Finally take $n = 8$ and

$$\mathbb{F}_{256} = \mathbb{F}_2[t]/(m(t)), \qquad m(t) = t^8 + t^4 + t^3 + t + 1, \qquad a = t^5 + t^3 + t^2 + t.$$

The irreducibility criterion for m is certified by $t^{256} \equiv t \pmod{m}$ and $\gcd(m, t^{16} + t) = 1$; these are polynomial identities over $\mathbb{F}_2$. Direct reduction modulo m gives

$$a^{85} = (1 + a)^{85} = 1, \qquad a^5 = t^7 + t^5 + t^3 + t + 1 \neq 1, \qquad a^{17} = t^7 + t^6 + t^5 + t^3 + t^2 + 1 \neq 1.$$

Since $85 = 5 \cdot 17$, these identities show that a has exact order 85 and $1 + a \in C_{85}$. Thus $a \in \mathcal{R}_{85}$; $f(x) = ax + c$ realizes the group for every $c \neq 0$ by Corollary 5.3. This closes all the even dimensions claimed. □

The square-root estimate above is classical; the results specialize it to the subgroup generated by the linked slopes a and $1 + a$. The sufficient bound is not a general necessary condition for realizability, and no priority claim is made for the character-sum method or these particular applications.

5.1 Quantitative realization and the quarter-power obstruction

The divisor sieve yields a main term as well as positivity. The next two results allow the subgroup index to grow with the field size.

Theorem 5.9 (Quantitative slope counts)**.** *Let $q = 2^n$, $d > 1$ divide $q - 1$, and $k = (q - 1)/d$. Set*

$$\theta_j(d) = \prod_{p \mid d} (1 - p^{-j}) \quad (j = 1, 2), \qquad S_{k,d} = \sum_{h \mid \mathrm{rad}(d)} b_{kh}, \qquad b_\ell = \frac{(\ell - 1)(\ell - 2)}{\ell^2}.$$

Here $\mathrm{rad}(d)$ *is the product of its distinct prime divisors. Then*

$$|\mathcal{R}_d| = \frac{q + 1}{k^2} \theta_2(d) - \frac{3}{k} \theta_1(d) + E_{q,k,d}, \qquad |E_{q,k,d}| \leq S_{k,d} \sqrt{q} \leq 2^{\omega(d)} \sqrt{q}. \tag{8}$$

The sharper error coefficient is

$$S_{k,d} = 2^{\omega(d)} - \frac{3}{k} \prod_{p \mid d} (1 + p^{-1}) + \frac{2}{k^2} \prod_{p \mid d} (1 + p^{-2}). \tag{9}$$

For fixed k, as n tends to infinity through admissible dimensions,

$$|\mathcal{R}_d| = \frac{q}{k^2} \theta_2(d) + O_k(q^{1/2 + o(1)}), \qquad \frac{|\mathcal{R}_d|}{T_d} = \theta_2(d) + O_k(q^{-1/2 + o(1)}). \tag{10}$$

In particular $\theta_2(d) > 3/4$: asymptotically at least three quarters of the slopes satisfying the two subgroup-membership conditions generate the specified complement jointly.

Proof. Write $T_{d/h} = (q+1-3kh)/(kh)^2 + \varepsilon_h$ in (2). Proposition 5.6 gives $|\varepsilon_h| \le b_{kh}\sqrt{q}$. Only squarefree h contribute. The identities

$$\sum_{h \mid d} \frac{\mu(h)}{h^j} = \theta_j(d), \qquad \sum_{h \mid \mathrm{rad}(d)} h^{-j} = \prod_{p \mid d}(1+p^{-j})$$

give (8) and, since $b_\ell = 1 - 3/\ell + 2/\ell^2$, (9). All these coefficients are nonnegative and at most one for integer $\ell \ge 1$.

For each $\epsilon > 0$, choose a fixed cutoff beyond which $2 \le p^\epsilon$. The bounded contribution of smaller primes can be absorbed into a constant, so $2^{\omega(d)} \le C_\epsilon d^\epsilon$. Consequently $2^{\omega(d)} = q^{o(1)}$. The first asymptotic follows from (8); the omitted terms are bounded for fixed k. Also $T_d = q/k^2 + O_k(\sqrt{q})$ by (4), which gives the ratio assertion. Finally $\prod(1-x_i) \ge 1 - \sum x_i$ for $0 \le x_i \le 1$, and the odd-prime bound in Proposition 5.6 is strictly less than $1/4$. Thus $\theta_2(d) > 3/4$. □

Corollary 5.10 (A uniform range of growing indices)**.** *Let $n \ge 16$, $q = 2^n$, and $k > 1$ divide $q-1$. If*

$$k \le \frac{q^{1/4}}{\sqrt{n}}, \tag{11}$$

then $H_{(q-1)/k}$ is a realizable proper primitive affine companion group.

Proof. Put $d = (q-1)/k$. Since $k < \sqrt{q}+1$, we have $d > \sqrt{q}-1$, so the field-degree argument in Theorem 5.7 gives $\mathrm{ord}_d(2) = n$. The index is odd and at least three; hence $(3k-1)/k^2 < 1$. Using $\omega(d) < 2n/3$ and $b_k \le 1$ gives

$$L_{q,k,d} > \left(\frac{n}{12} - 1\right)\sqrt{q} - 1 > 0 \qquad (n \ge 16).$$

Theorem 5.7 proves the assertion. □

The exponent $1/4$ has a structural obstruction. It is not merely an artifact of bounding each Jacobi sum separately.

Theorem 5.11 (Norm descent into a proper subfield)**.** *Let $1 \le s < r$, $s \mid r$, $u \mid 2^s - 1$, and work in $L = \mathbb{F}_{2^{2r}}$. Put*

$$d = u(2^r+1), \qquad g = \begin{cases} u, & r/s \text{ even}, \\ u(2^s+1), & r/s \text{ odd}. \end{cases}$$

The two membership conditions descend exactly:

$$\{a \in C_d : 1 + a \in C_d\} = \{a \in C_g \subset \mathbb{F}_{2^{2s}}^\times : 1 + a \in C_g\}. \tag{12}$$

Thus $T_d = T_g$, $\mathcal{R}_d = \varnothing$, and $H_d = L^+ \rtimes C_d$ is primitive but cannot be generated by an affine complete mapping and its prescribed companion. The nonrealizability also holds for every linear conjugate of H_d.

Proof. Set $Q = 2^r$, $K = \mathbb{F}_Q$, and $E = \mathbb{F}_{2^s} \subset K$. The norm $N_{L/K}(x) = x^{Q+1}$ has kernel C_{Q+1} and maps C_d onto $C_u \subset E^\times$: its image has order $d/\gcd(d, Q+1) = u$, since $\gcd(u, Q+1) = 1$. If $a, 1+a \in C_d$, put $U = N(a)$ and $W = N(1+a)$. Both lie in E, and

$$a + a^Q = 1 + U + W \in E, \qquad a^2 + (1+U+W)a + U = 0.$$

Thus a has degree at most two over E and belongs to $\mathbb{F}_{2^{2s}} \subset L$. The same holds for $1+a$.

Write $t = 2^s$ and $m = r/s$. Cyclic subgroup intersection gives $C_d \cap \mathbb{F}_{t^2}^\times = C_{\gcd(d,t^2-1)}$. If m is even, $t^m + 1 \equiv 2 \pmod{t^2-1}$, so this gcd is u. If m is odd, $t^m + 1 \equiv t + 1 \pmod{t^2-1}$; since $u \mid t-1$ and $\gcd(t-1, t+1) = 1$, the gcd is $u(t+1)$. This proves one inclusion in (12); the other follows from $C_g \subset C_d$. In both cases $g < d$, so every pair in the left-hand set generates a proper subgroup of C_d. Therefore $\mathcal{R}_d = \varnothing$.

The group C_d contains C_{2^r+1}, whose action on L is irreducible by the field-degree proof in Theorem 5.5. Hence H_d is primitive. Corollary 5.3 excludes scalar realizations, and every affine permutation in the specified group is scalar. Linear-conjugacy equivariance of F excludes realizations of its linear conjugates as well. □

Corollary 5.12 (The quarter-power barrier)**.** *For every $s \geq 1$, put $t = 2^s$, $q = t^4$, and*

$$k = t + 1 = q^{1/4} + 1, \qquad d = (t-1)(t^2+1) = \frac{q-1}{k}.$$

Then H_d is a proper primitive nonrealizable affine companion group, and its membership pairs are exactly

$$\{a \in C_d : 1 + a \in C_d\} = \mathbb{F}_t \setminus \{0, 1\}. \qquad T_d = t - 2, \quad |\mathcal{R}_d| = 0. \tag{13}$$

Consequently no exponent $\beta > 1/4$ can give universal eventual realizability for all admissible indices $1 < k \leq q^\beta$. Even the range $1 < k \leq Cq^{1/4}$ fails for every fixed $C > 1$.

Proof. Apply Theorem 5.11 with $r = 2s$ and $u = t - 1$. Here $g = t - 1$ and $C_g = \mathbb{F}_t^\times$, yielding (13). The indices are proper and $k/q^{1/4} = 1 + 1/t \to 1$. Hence these infinitely many exceptions eventually lie in either range in the statement. □

Together, Corollaries 5.10 and 5.12 determine the exponent $1/4$ for a universal growing-index guarantee. They leave a factor of $\sqrt{n} = \sqrt{\log_2 q}$ between the sufficient range and the obstructed family. In particular, the exact boundary $k \leq q^{1/4}$ has not been decided by these arguments.

5.2 Primitive classification and generating triangles

Proposition 5.13 (Primitive inverse classification and linear conjugacy)**.** *Up to* $\mathrm{GL}(n, 2)$ *conjugacy, the primitive groups generated by an affine complete mapping and its prescribed companion are exactly*

$$H_d = \mathbb{F}_{2^n}^+ \rtimes C_d, \qquad d \mid 2^n - 1, \quad \mathrm{ord}_d(2) = n, \quad \mathcal{R}_d \neq \varnothing.$$

Distinct admissible d give distinct group-conjugacy classes. For each such d, the normalizer in $\mathrm{GL}(n, 2)$ *has order $n(2^n - 1)$ and is the semilinear scalar group $\Gamma\mathrm{L}(1, 2^n)$. The affine complete mappings realizing a linear conjugate of H_d form exactly $|\mathcal{R}_d|/n$ linear-conjugacy classes, each of size $|\mathrm{GL}(n, 2)|$.*

Proof. Corollary 5.2 makes the linear part A of every primitive realization irreducible and its translation nonzero. The field $\mathbb{F}_2[A]$ then identifies V with $\mathbb{F}_{2^n}$; the complement is generated by a and $1 + a$ and has some order d. The characterization follows from Corollary 5.3. Conversely that corollary realizes every listed H_d. Different orders $2^n d$ distinguish the group classes.

A linear transformation normalizes H_d exactly when it normalizes its stabilizer C_d of zero. Since $\mathbb{F}_2[C_d] = \mathbb{F}_{2^n}$, conjugation by such a transformation induces an automorphism of this field, hence a power of Frobenius. After removing that power, the transformation centralizes all scalar multiplications and is itself scalar. Conversely every map $x \mapsto bx^{2^i}$ normalizes C_d. The normalizer therefore has the asserted order.

For maps realizing the specified H_d, normalizer conjugation sends (a, c) to (a^{2^i}, bc^{2^i}). Scalar transformations act transitively on $c \neq 0$. Every $a \in \mathcal{R}_d$ has degree n over $\mathbb{F}_2$: a proper subfield containing a would contain $1 + a$ and all of C_d. Thus its Frobenius orbit has length n, giving $|\mathcal{R}_d|/n$ normalizer orbits. Conjugacy between two maps with group H_d must normalize H_d, so these are also the full linear-conjugacy classes when group conjugates are included. Finally, a linear transformation centralizing $f(x) = ax + c$ centralizes the irreducible multiplier a, so is scalar; fixing $c \neq 0$ forces it to be the identity. All full linear-conjugacy classes consequently have size $|\mathrm{GL}(n, 2)|$. □

For an explicit polynomial encoding, set

$$G_e(t) = \gcd(t^e + 1, (t + 1)^e + 1), \qquad P_d(t) = \prod_{e|d} G_e(t)^{\mu(d/e)}.$$

Although written as a quotient, P_d is a monic squarefree polynomial over $\mathbb{F}_2$ with root set $\mathcal{R}_d$. Indeed the exponent of a root whose pair generates C_h is $\sum_{h|e|d} \mu(d/e)$, which is one if $h = d$ and zero otherwise. When $\mathrm{ord}_d(2) = n$, all irreducible factors of P_d have degree n and encode the classes in Proposition 5.13. For example the index-three counts in dimensions $8, 10, 12, 14$ give $3, 9, 33, 123$ map classes, respectively.

The graph interpretation clarifies the distinction between subgroup membership and joint generation. Write

$$\Gamma(q, d) = \mathrm{Cay}(\mathbb{F}_q^+, C_d), \qquad x \sim y \iff y - x \in C_d.$$

This is the usual generalized Paley graph (including the complete graph when $d = q - 1$). Its connectivity and ordinary triangle counts are classical [6, 7]. For a triangle with ordered vertices (x, y, z), define its ratio subgroup to be

$$K(\{x, y, z\}) = \langle a, 1 + a \rangle, \qquad a = \frac{z - x}{y - x}.$$

This subgroup is independent of the vertex ordering. We call the triangle *generating* when its ratio subgroup is C_d. The definition uses normalized ratios, not the subgroup generated by the unnormalized edge differences.

Proposition 5.14 (Generating triangles and companion realization)**.** *For $d > 1$ dividing $q - 1$, $\Gamma(q, d)$ is connected exactly when $\mathrm{ord}_d(2) = n$. Each edge belongs to T_d ordinary triangles and $|\mathcal{R}_d|$ generating triangles. The total number of generating triangles is $qd|\mathcal{R}_d|/6$. Thus H_d is realizable exactly when the graph is connected and has a generating triangle. If $d > 3$, the action of H_d on generating triangles is free, with $|\mathcal{R}_d|/6$ orbits; in particular $6 \mid |\mathcal{R}_d|$.*

Proof. The additive span of C_d is closed under products, contains one, and is a finite subfield. It is therefore $\mathbb{F}_2[C_d] = \mathbb{F}_{2^{\mathrm{ord}_d(2)}}$. The components are its additive cosets, proving the connectivity assertion (compare [6, Theorem 2.2]).

For the edge $\{0, 1\}$, its third vertices are precisely the a with $a, 1 + a \in C_d$, and the generating ones are precisely $a \in \mathcal{R}_d$. Reordering vertices replaces a by one of

$$a, \quad 1 + a, \quad a^{-1}, \quad (1 + a)^{-1}, \quad \frac{a}{1 + a}, \quad \frac{1 + a}{a};$$

these all generate the same subgroup together with one plus themselves. Translations and multiplication by C_d preserve the ratios and act transitively on edges. Counting the $qd/2$ edges and three edges per triangle gives the total. The realization equivalence now follows from Corollary 5.3.

For $d > 3$ the six transforms of $a \in \mathcal{R}_d$ are distinct. A transposition can fix only an excluded value 0 or 1, whereas a three-cycle can fix only $a^2 + a + 1 = 0$, whose pair generates C_3. Any affine stabilizer of a generating triangle induces a vertex permutation and would make the corresponding transform fix a. It must therefore fix the vertices individually, forcing its multiplier to be one and its translation to be zero. The action is free. Each orbit has size qd, giving the orbit count and divisibility. No claim about the full graph automorphism group is required here. □

In the family of Corollary 5.12, the graph is connected and every edge has exactly $t-2$ third vertices. For an edge $\{x, y\}$ they are $x + (y - x)a$ with $a \in \mathbb{F}_t \setminus \{0, 1\}$. Thus all its triangles lie in the clique $x + (y - x)\mathbb{F}_t$, and none is generating. These graphs are the already studied $(t+1)$-power graphs $\Gamma_{t,4}(1)$ of Podestá and Videla [8]; their ordinary intersection parameter $t-2$ follows also from [8, Theorem 5.2]. The descent proof supplies the exact subfield location and its companion-group obstruction. Connectedness and ordinary triangles alone do not guarantee realization. The graph family, character-sum method, and ordinary triangle counts are not claimed as new, and priority of the companion-specific applications remains to be established.

5.3 Realization at the inner quarter-power boundary

The obstruction at $k = t + 1$ has a positive counterpart at $k = t - 1$. Throughout this subsection put

$$t = 2^s \ (s \geq 2), \quad K = \mathbb{F}_t, \quad E = \mathbb{F}_{t^2}, \quad L = \mathbb{F}_{t^4}, \quad A = t + 1, \quad B = t^2 + 1, \quad d = AB.$$

Then $\gcd(A, B) = 1$ and $C_d = \ker N_{L/K}$. Write $U = \ker N_{E/K} = C_A$ and $V = \ker N_{L/E} = C_B$. The next results prove the stronger assertion that some $a \in C_d$ has order d and satisfies $1 + a \in C_d$.

Proposition 5.15 (Quadratic parametrization)**.** *The elements $a \in L \setminus E$ with $a, 1 + a \in C_d$ correspond, two at a time, to the ordered pairs $(b, w) \in U^2$ satisfying*

$$c = 1 + b + w \neq 0, \qquad \mathrm{Tr}_{E/\mathbb{F}_2}(b/c^2) = 1.$$

The two elements are the roots of $X^2 + cX + b$, and $N_{L/E}(a) = b$, $N_{L/E}(1+a) = w$. The remaining elements, in E, are precisely the two roots of $X^2 + X + 1$ when s is odd, and there are none when s is even. These remaining elements generate only C_3.

Proof. If $a \notin E$, write $b = a^{t^2+1}$ and $c = a + a^{t^2}$. Then $c \neq 0$, the minimal polynomial is $X^2 + cX + b$, and $(1+a)^{t^2+1} = 1 + b + c = w$. Norm transitivity puts b, w in U. After division by c^2, the quadratic is an Artin–Schreier equation; it is irreducible exactly when the displayed absolute trace is one. Conversely, that trace condition makes the quadratic irreducible over E, with both roots in $L \setminus E$. Their two relative norms are b, w, so norm transitivity gives $a, 1 + a \in C_d$. The relative norms determine the ordered pair uniquely.

For $a \in E$, the same norm conditions reduce to $a, 1 + a \in U$, since $N_{L/K}(a) = N_{E/K}(a)^2$. Thus $a^t = a^{-1}$ and $(1 + a)^t = (1 + a)^{-1}$. Substitution gives $(1 + a)(1 + a^{-1}) = 1$, or $a^2 + a + 1 = 0$. Such roots lie in U exactly when $3 \mid t + 1$, equivalently when s is odd. Their two norms satisfy $1 + b + w = 0$. In particular, the omitted $c = 0$ branch contributes no realizing slope. □

Lemma 5.16 (A genus-one norm fiber)**.** *Fix $b \in U$ with $b^3 \neq 1$ and put*

$$X_b(K) = \{a \in L : N_{L/E}(a) = b, \ N_{L/K}(1 + a) = 1\}.$$

This set is the full set of K-points of a smooth projective geometrically connected curve C_b/K of genus one. Over $\overline{K}$ the affine part has equation

$$(x+1)(x+b)(y+1)(y+b^{-1}) = xy \qquad (x, y \neq 0). \tag{14}$$

Its completion in $\mathbb{P}^1 \times \mathbb{P}^1$ has eight boundary points, none rational over K. Consequently

$$\left|\#X_b(K) - (t+1)\right| \leq 2\sqrt{t}.$$

Proof. The relative-norm equation defines a variety over K by restriction of scalars. In the four split coordinates of $\mathrm{Res}_{L/K}\,\mathbb{G}_m$, write $(a_0, a_1, a_2, a_3) = (x, y, b/x, b^{-1}/y)$. The remaining norm equation is exactly (14). Let $P = (x+1)(x+b)$ and $Q = (y+1)(y+b^{-1})$. At a singular point in the torus, $PQ = xy \neq 0$ and

$$(b+1)Q + y = 0, \qquad (b^{-1}+1)P + x = 0.$$

Multiplication would force $b + b^{-1} = 1$, hence $b^3 = 1$. This contradicts the hypothesis $b^3 \neq 1$. The boundary points are

$$D_x = \{(0,1), (0,b^{-1}), (\infty,1), (\infty,b^{-1})\},$$
$$D_y = \{(1,0), (b,0), (1,\infty), (b,\infty)\}.$$

They are distinct and smooth: in the appropriate affine charts the derivative along the finite coordinate is a nonzero multiple of $1+b$ or $1+b^{-1}$. No corner of the square is on the curve. The smooth bidegree-$(2,2)$ curve is geometrically connected: distinct components with zero intersection would all have to be fibers of the same projection, contrary to the total bidegree. It is therefore geometrically irreducible and has genus $(2-1)(2-1) = 1$. Its smooth completion descends to K.

The t-power Frobenius on these coordinates is $(x, y) \mapsto (b/y^t, x^t)$. A rational point satisfies $y = x^t$, which excludes all eight boundary points. Hence $C_b(K) = X_b(K)$, with the stated interpretation in L. The final bound is the Hasse–Weil bound. □

Lemma 5.17 (Uniform character bound on a norm fiber)**.** *Let b be as in Lemma 5.16, and let $u \in U$ be the unique square root of b. For every nontrivial multiplicative character χ of V, one has*

$$\left|\sum_{a \in X_b(K)} \chi(a/u)\right| \leq 8\sqrt{t}. \tag{15}$$

Proof. We use the classical Weil bound for a geometrically nontrivial finite-order rank-one character on a curve: on a genus-g curve, tame ramification at r geometric points gives the bound $(2g-2+r)\sqrt{t}$, when the character is nontrivial at every one of those points. This is the curve Artin-character estimate discussed in [10, §4]; the Lang-torsor interpretation of finite-field multiplicative characters is also used in [9]. For completeness, the dimension underlying this estimate is $2g-2+r$: compactly supported cohomology in degrees zero and two vanishes, and the tame Euler-characteristic formula gives this dimension in degree one. The trace formula and the Weil theorem for curves bound each eigenvalue by $\sqrt{t}$. For finite-order characters this is a character component of a finite cover of the curve. We now check all these hypotheses here.

Let $\mathcal{T} = \mathrm{Res}_{E/K}(\ker N_{L/E})$ denote the two-dimensional norm torus. It has $\mathcal{T}(K) = V$, and $a \mapsto a/u$ maps X_b to $\mathcal{T}$, since $N_{L/E}(u) = u^2 = b$. Over $\overline{K}$ use coordinates (z_0, z_1) on the torus; its Frobenius is

$$F_{\mathcal{T}}(z_0, z_1) = (z_1^{-t}, z_0^t).$$

The Lang isogeny $z \mapsto F_{\mathcal{T}}(z)/z$ has constant kernel V. Pulling it back along $a \mapsto a/u$, and writing $r_0 = x/u$, $r_1 = y/u^t$, gives

$$\frac{z_1^{-t}}{z_0} = r_0, \qquad \frac{z_0^t}{z_1} = r_1, \qquad \text{hence} \qquad z_0^B = \frac{r_1^t}{r_0}.$$

Thus this pullback is geometrically a Kummer cover with defining function $h = (y/u^t)^t/(x/u) = y^t/x$, since $u^{t^2} = u$. For a character of exact order $m \mid B$, its quotient cover has equation $z^m = h$ over $\overline{K}$. This geometric Kummer presentation must be distinguished from the K-structure of the quotient cover: the latter is inherited from the pullback of the K-defined Lang isogeny, with its corresponding Frobenius descent datum. In particular, h need not belong to the function field $K(C_b)$ in these split coordinates. All statements about geometric ramification and geometric connectedness are checked after base change to $\overline{K}$, whereas the character sum is taken using the descended K-cover.

The Frobenius descent can be made explicit, so that the character trace is not merely inferred from a convention. On geometric points, the K-Frobenius on the base and on the pullback Lang cover is

$$\begin{aligned} F_{C_b}(x, y) &= \left(\frac{b}{y^t}, x^t\right), \\ F_Y(x, y, z_0) &= \left(\frac{b}{y^t}, x^t, \frac{x}{u} z_0\right). \end{aligned} \tag{16}$$

Indeed, the two Lang equations give $z_1^{-t} = r_0 z_0$, while $F_{\mathcal{T}}$ acts on the first split coordinate as $z_0 \mapsto z_1^{-t}$. The displayed formula is also compatible with the Kummer equation, since $u^B = b$ and

$$h\left(\frac{b}{y^t}, x^t\right) = \frac{x^{t^2} y^t}{b} = \left(\frac{x}{u}\right)^B h(x, y).$$

For a K-rational point, $(x, y) = (a, a^t)$ with $a^B = b$; thus Frobenius fixes the base point and acts on its geometric Lang fiber by $z_0 \mapsto (a/u) z_0$. This is precisely the deck transformation indexed by $a/u \in V$. In the character eigensheaf associated with χ, its Frobenius trace is therefore $\chi(a/u)$, or $\chi^{-1}(a/u)$ if the inverse Frobenius convention is used. The latter choice is immaterial for the asserted bound. In particular, this computation verifies the Frobenius trace identification for the actual K-descent of the cover, rather than for the bare geometric Kummer equation.

The zeros and poles of x are simple and supported on D_x; those of y are simple and supported on the disjoint set D_y. The valuations can be read point by point:

points	$\mathrm{ord}_P(h)$
$(0, 1),\ (0, b^{-1})$	-1
$(\infty, 1),\ (\infty, b^{-1})$	1
$(1, 0),\ (b, 0)$	t
$(1, \infty),\ (b, \infty)$	$-t$

Thus the eight nonzero valuations are $1, -1, t, -t$, each twice. Since m is odd and $\gcd(m, t) = 1$, the character is tamely and nontrivially ramified at all eight points. A valuation of absolute value one also makes this cover geometrically connected, excluding a constant-field character or a geometrically trivial pullback. The conductor degree is consequently exactly eight, independently of m, t, b. Lemma 5.16 and the curve bound now give $(2 \cdot 1 - 2 + 8)\sqrt{t} = 8\sqrt{t}$. □

Proposition 5.18 (Four-torsion symmetry of a norm fiber)**.** *In the setting of Lemma 5.16, let $u \in U$ satisfy $u^2 = b$. On the geometric (x, y)-model of C_b set*

$$\rho(x, y) = \left(\frac{1}{y}, \frac{x}{b}\right), \qquad h = \frac{(y/u^t)^t}{x/u} = \frac{y^t}{x}.$$

Then ρ descends to a K-automorphism of order four. After choosing a K-rational origin, it is translation by a K-rational point of exact order four. The eight boundary points form exactly two ρ-orbits of length four:

$$(0, 1) \longmapsto (1, 0) \longmapsto (\infty, b^{-1}) \longmapsto (b, \infty) \longmapsto (0, 1),$$
$$(0, b^{-1}) \longmapsto (b, 0) \longmapsto (\infty, 1) \longmapsto (1, \infty) \longmapsto (0, b^{-1}).$$

Moreover, as identities in the geometric function field,

$$\rho^* h = (uy)^B h^{-t}, \qquad (\rho^2)^* h = h^{-1}. \tag{17}$$

For every character χ of V, the sums in Lemma 5.17 satisfy

$$S_b(\chi) := \sum_{a \in X_b(K)} \chi(a/u) = S_b(\chi^{-t}) = S_b(\chi^{-1}) \in \mathbb{R}.$$

Every nontrivial character belongs to an orbit of length four under $\chi \mapsto \chi^{-t}$. If N_b^{prim} denotes the number of $a \in X_b(K)$ with $\mathrm{ord}(a) = d$, then $4 \mid \#X_b(K)$ and $4 \mid N_b^{\mathrm{prim}}$.

Proof. The field automorphism $\sigma : a \mapsto a^t$ of L/K extends as $\sigma \otimes 1$ to an R-linear automorphism of $L \otimes_K R$ for every K-algebra R; composing this extension with inversion gives a K-automorphism of the ambient restriction-of-scalars torus. On K-points its action is $a \mapsto (a^t)^{-1}$. Here this notation refers to the Galois automorphism σ, not the relative Frobenius morphism on the torus. The functorial norm identities

$$N_{L/E}((\sigma a)^{-1}) = b^{-t} = b, \qquad N_{L/K}(1 + (\sigma a)^{-1}) = \frac{N_{L/K}(1 + a)}{N_{L/K}(a)} = 1$$

show that it preserves both defining conditions, since $N_{L/K}(a) = N_{E/K}(b) = 1$. It gives the displayed ρ in split coordinates. Equivalently, the split-coordinate K-Frobenius descent is $(x, y) \mapsto (b/y^t, x^t)$, with which ρ commutes. Direct iteration gives

$$\rho^2(x, y) = (b/x, b^{-1}/y), \qquad \rho^4 = 1.$$

The eight boundary points are paired without fixed points by ρ^2. Any remaining geometric fixed point would have $x^2 = b$, $y^2 = b^{-1}$, hence $(x, y) = (u, u^{-1})$. Substitution in the equation of C_b forces $b + b^{-1} = 1$, equivalently $u^2 + u + 1 = 0$, whence $b^3 = 1$, contrary to $b^3 \neq 1$. Thus ρ^2, and hence ρ, has no geometric fixed point. The bound $\#C_b(K) \geq t + 1 - 2\sqrt{t} > 0$ supplies a rational origin. After choosing a rational origin, any automorphism of a genus-one curve is a translation followed by an origin-preserving automorphism. If the latter is nontrivial, its difference from the identity is a nonzero, hence surjective, elliptic-curve endomorphism, so the composite has a geometric fixed point. Consequently ρ is a translation, necessarily by a K-rational point of exact order four.

The two boundary cycles follow by substitution in the rational formula, interpreted on $\mathbb{P}^1 \times \mathbb{P}^1$. Since $u^{t+1} = 1$, $u^B = u^2 = b$ and $b^t = b^{-1}$,

$$\rho^* h = \frac{(x/b)^t}{1/y} = b x^t y = (uy)^B \left(\frac{y^t}{x}\right)^{-t},$$

while the formula for ρ^2 yields $(\rho^2)^* h = h^{-1}$. For $a = uv$ with $v \in V$, ρ sends a to uv^{-t}. It therefore permutes $X_b(K)$ and preserves $\mathrm{ord}(a)$; changing variables in the character sum proves the stated equalities. Finally $B = t^2 + 1$ is odd, $(-t)^2 \equiv -1 \pmod{B}$. For a character of order $m > 1$, an orbit of length one or two would imply $m \mid 2$, which is impossible since $m \mid B$ is odd. Thus every nontrivial character has an orbit of length four. Translation by a point of exact order four acts freely, proving both divisibilities. □

These symmetries do not by themselves reduce the bound in Lemma 5.17: grouping the characters into four-element orbits leaves the total absolute sieve coefficients unchanged. The estimate used below remains $8\sqrt{t}$.

Theorem 5.19 (Primitive slopes in the inner critical family)**.** *Put $\theta(B) = \varphi(B)/B$ and $W(B) = 2^{\omega(B)}$. For each $b \in U$ of order A, let*

$$N_b^{\mathrm{prim}} = \#\{a \in X_b(K) : \mathrm{ord}(a) = d\}.$$

Then

$$\left|N_b^{\mathrm{prim}} - \theta(B)(t+1)\right| \le \theta(B)(8W(B) - 6)\sqrt{t}. \tag{18}$$

In particular, with $N_d^{\mathrm{prim}} = \#\{a \in C_d : \mathrm{ord}(a) = d,\ 1 + a \in C_d\}$,

$$\left|N_d^{\mathrm{prim}} - \varphi(A)\theta(B)(t+1)\right| \le \varphi(A)\theta(B)(8W(B) - 6)\sqrt{t}. \tag{19}$$

Thus $N_d^{\mathrm{prim}} \sim \varphi(A)\theta(B)t$ as $s \to \infty$.

Proof. Every $a \in X_b(K)$ has a unique expression $a = uv$ with $v \in V$, where u is the square root of b in U. Since $\mathrm{ord}(u) = A$ and $\gcd(A, B) = 1$, it has order d exactly when v has order B. The usual character indicator in a cyclic group of order B is

$$\mathbf{1}_{\mathrm{ord}(v)=B} = \theta(B) \sum_{m \mid B} \frac{\mu(m)}{\varphi(m)} \sum_{\mathrm{ord}(\chi)=m} \chi(v).$$

Indeed the sum factors over primes $p \mid B$: the corresponding factor is zero on V^p and $p/(p-1)$ off V^p, which cancels the factor $(p-1)/p$ in $\theta(B)$. The trivial-character term is $\theta(B)\#X_b(K)$. For each nontrivial squarefree divisor m the sum of the absolute values of its coefficients is one. There are $W(B) - 1$ such divisors. Lemmas 5.16 and 5.17 therefore give an error of at most $\theta(B)(2 + 8(W(B) - 1))\sqrt{t}$, proving (18).

An element of order d has relative norm of order A, because $N_{L/E}(a) = a^B$. Hence these $\varphi(A)$ norm fibers partition the elements counted by N_d^{prim}. Summing proves (19). Finally $2^{\omega(N)} = N^{o(1)}$: for any $\epsilon > 0$, all but finitely many primes satisfy $2 \le p^\epsilon$, and the remaining primes contribute only a constant. Since $B \asymp t^2$, the relative error tends to zero. □

Theorem 5.20 (All-dimensional realization at index $t-1$)**.** *For every $s \ge 2$, there exists $a \in \mathbb{F}_{2^{4s}}$ with*

$$\mathrm{ord}(a) = (2^s + 1)(2^{2s} + 1), \qquad (1 + a)^{(2^s+1)(2^{2s}+1)} = 1.$$

Consequently $\mathcal{R}_d \neq \varnothing$ at $k = t-1$ for every such s. The proper primitive group $L^+ \rtimes C_d$ is generated by $f(x) = ax + c$ and $F(f)$ for every $c \in L^\times$.

Proof. Every prime divisor p of $B = t^2 + 1$ is 1 modulo 4, since $t^2 \equiv -1 \pmod{p}$. If $p \geq 256$, then $2 \leq p^{1/8}$. For the smaller primes a direct integer product gives

$$\prod_{\substack{p<256 \text{ prime} \\ p\equiv 1 \pmod 4}} \frac{256}{p} < 32^8.$$

The 24 primes in this product and the exact integer comparison are supplied with the finite verifier in Appendix A. It follows for every B under consideration that

$$W(B) < 32B^{1/8}.$$

For $s \geq 33$, $t^2 \geq 2^{66}$ and $t^2+1 < 2t^2$ imply $t^4 > 2^{64}(t^2+1)$. Equivalently, $\sqrt{t} > 256B^{1/8} > 8W(B)$. The lower bound in (18) is then positive for every primitive $b \in U$.

For $2 \leq s \leq 32$, Appendix A gives explicit field towers $E = \mathbb{F}_2[X]/(M_s)$ and $L = E[Z]/(Z^2 + cZ + b)$, in which $a = Z$ is a certificate. Irreducibility, both norm conditions and the exact order d are verified by deterministic finite arithmetic, with complete factorizations and recursively certified primes. These 31 cases exhaust the remaining range; they are not an extrapolation.

Finally $d > t^2 - 1$ implies $\mathbb{F}_2[a] = L$: every proper subfield of L has at most t^2 elements. Thus C_d acts irreducibly, while $k = t - 1 > 1$ makes it a proper multiplier group. The affine conclusion follows from Corollary 5.3. □

The two adjacent indices $t-1$ and $t+1$ therefore have opposite realization behavior for every $s \geq 2$. This does not settle the universal question for all admissible $k \leq q^{1/4}$. Also, N_d^{prim} counts primitive individual slopes; it is a subset count within $|\mathcal{R}_d|$. The joint and double full-order counts below require the mixed-character estimate. Ordinary intersections for index $t-1$ belong to cyclotomic number theory, including the general norm-based treatment of Kudo and Nogata [18]. The additional exact-order condition is the focus here. The classical Weil bound, Lang construction, and primitive-character sieve are attributed above; priority for this particular realization theorem is not asserted.

Mixed characters and joint densities. We now retain all three varying character components on a norm fiber. For $a \in X_b(K)$ put

$$w = (1+a)^B \in U, \qquad v = a^A \in V, \qquad v_+ = (1+a)^A \in V. \tag{20}$$

Lemma (Character decomposition)**.** *The map $C_d \longrightarrow U \times V$, $z \mapsto (z^B, z^A)$, is an isomorphism. Every pair $\xi, \zeta \in \widehat{C_d}$ has a unique decomposition*

$$\xi(z) = \alpha(z^B)\chi(z^A), \qquad \zeta(z) = \eta(z^B)\psi(z^A), \quad \alpha, \eta \in \widehat{U}, \quad \chi, \psi \in \widehat{V}.$$

Define the two-character sum and the norm-character exception set by

$$\begin{aligned} S_b(\xi,\zeta) &= \sum_{a \in X_b(K)} \xi(a)\zeta(1+a), \\ \mathcal{E}_{\mathrm{norm}} &= \{(\alpha \circ (z \mapsto z^B), 1) : \alpha \in \widehat{U}\}. \end{aligned} \tag{21}$$

For every $b \in U$ the sum is

$$S_b(\xi,\zeta) = \alpha(b) \sum_{a \in X_b(K)} \eta(w)\chi(v)\psi(v_+).$$

Moreover $(\xi, \zeta) \in \mathcal{E}_{\mathrm{norm}}$ if and only if $\eta = \chi = \psi = 1$.

Proof. The two powers have orders dividing A and B, respectively. An element in the kernel has both its A-th and B-th powers equal to one, hence is one because $\gcd(A, B) = 1$. The source and target have the same order AB, so the map is an isomorphism. Taking characters gives the unique decomposition. On the fiber $a^B = b$; substitution gives the identity for the sum. Uniqueness also proves the stated equivalence for the exception set. □

Proposition 5.21 (Mixed characters on a norm fiber)**.** *Let $b \in U$ satisfy $b^3 \neq 1$. For characters $\eta \in \widehat{U}$ and $\chi, \psi \in \widehat{V}$, not all trivial,*

$$\left| \sum_{a \in X_b(K)} \eta(w)\chi(v)\psi(v_+) \right| \leq 8\sqrt{t}. \tag{22}$$

The product Lang cover used for this sum is geometrically connected, and hence its smooth projective normalization is geometrically irreducible. Every character quotient is tame, with conductor degree at most eight.

Proof. Let $C_b^\circ = C_b \setminus (D_x \cup D_y)$, with the eight points of Lemma 5.16. This open curve has $C_b^\circ(K) = X_b(K)$. Write $\sigma \in \mathrm{Gal}(L/K)$ for $z \mapsto z^t$, extended as $\sigma \otimes 1$ on every $L \otimes_K R$. The algebraic map

$$\nu(z) = z\,\sigma(z)$$

sends the norm-one torus $\ker N_{L/K}$ to $\mathcal{T} = \mathrm{Res}_{E/K}(\ker N_{L/E})$:

$$N_{L/E}(\nu(z)) = N_{L/E}(z)\,\sigma(N_{L/E}(z)) = N_{L/K}(z) = 1.$$

Its value on K-points is z^A. This construction is needed for the varying norm of $1+a$: exponentiation by A on geometric points would give relative norm w^A, which is not the constant function one.

Set $w_0 = (x+1)(x+b)/x$. The split coordinates of the three K-defined maps to the norm tori are

	first coordinate	second coordinate
$N_{L/E}(1+a)$	w_0	w_0^{-1}
$\nu(a)$	xy	by/x
$\nu(1+a)$	$(x+1)(y+1)$	$(y+1)(x+b)/x$

The first row is a map to $\mathcal{U} = \ker N_{E/K}$, and $\mathcal{U}(K) = U$, $\mathcal{T}(K) = V$. On $\mathcal{U}$ the Frobenius is $z \mapsto z^{-t}$; its Lang equation is $z^{-A} = w_0$. On $\mathcal{T}$ the Frobenius is $(z_0, z_1) \mapsto (z_1^{-t}, z_0^t)$; the Lang equations $z_1^{-t}/z_0 = r_0$, $z_0^t/z_1 = r_1$ give $z_0^B = r_1^t/r_0$. Consequently the product pullback is, over $\overline{K}$,

$$z_U^A = w_0^{-1}, \qquad z_1^B = g_1, \qquad z_2^B = g_2, \tag{23}$$

where

$$g_1 = b^{-1}\frac{y^{t-1}}{x^A}, \qquad g_2 = w_0^t\frac{(y+1)^{t-1}}{(x+1)^A}. \tag{24}$$

For the second formula, substitute $x + b = xw_0/(x+1)$ in r_1^t/r_0.

These geometric equations inherit their K-structure from the Lang pullbacks. More explicitly, above $F_{C_b}(x, y) = (b/y^t, x^t)$ their Frobenius sends the three displayed covering coordinates to

$$\big(w_0 z_U,\ xyz_1,\ (x+1)(y+1)z_2\big). \tag{25}$$

This follows from $F(z) = rz$ in the Lang equations. It is compatible with (23), since

$$w_0(F_{C_b}) = w_0^{-t}, \quad g_1(F_{C_b}) = (xy)^B g_1, \quad g_2(F_{C_b}) = ((x+1)(y+1))^B g_2.$$

At a rational point $(x, y) = (a, a^t)$, the three deck multipliers in (25) are precisely w, v, v_+. Thus the character component has trace $\eta(w)\chi(v)\psi(v_+)$; the inverse Frobenius convention simultaneously inverts the characters and gives the same bound.

We check geometric connectedness and the conductor directly. The divisors of w_0, g_1, g_2 are supported on the eight boundary points. Their valuations are

P	$\mathrm{ord}_P(w_0)$	$\mathrm{ord}_P(g_1)$	$\mathrm{ord}_P(g_2)$
$(0,1)$	-1	$-A$	-1
$(0,b^{-1})$	-1	$-A$	$-t$
$(\infty,1)$	-1	A	t
(∞,b^{-1})	-1	A	1
$(1,0)$	1	$t-1$	-1
$(b,0)$	1	$t-1$	t
$(1,\infty)$	1	$-(t-1)$	$-t$
(b,∞)	1	$-(t-1)$	1

(26)

Indeed x, y have simple zeros and poles on D_x, D_y; $x+1$ has simple zeros at $(1,0), (1,\infty)$ and simple poles at the two points with $x = \infty$, and the corresponding assertion holds for $y+1$. Substitution in (24) gives each entry of the table.

For every $m \mid B$, a relation $g_1^r g_2^s = f^m$ in $\overline{K}(C_b)^\times$ would imply, at $(0,1)$ and $(0,b^{-1})$,

$$m \mid Ar + s, \qquad m \mid Ar + ts.$$

Now $\gcd(B, t-1) = \gcd(t^2+1, t-1) = 1$ and $\gcd(A, B) = 1$. Subtracting forces $m \mid s$, and then $m \mid r$. The valuation -1 of w_0 likewise excludes every nontrivial Kummer power relation for an order dividing A. One can verify simultaneous independence in a common exponent:

$$w_0^{Be_0} g_1^{Ae_1} g_2^{Ae_2} \in \overline{K}(C_b)^{\times AB} \quad \Longrightarrow \quad A \mid e_0,\ B \mid e_1,\ B \mid e_2.$$

Reduction of the first two valuations modulo A gives $A \mid e_0$; their difference modulo B gives $B \mid e_2$, and either valuation then gives $B \mid e_1$. This is independence of the relevant functions modulo powers in $\overline{K}(C_b)^\times$, verified by their local valuations.

At $P_1 = (0,1)$ and $P_2 = (0,b^{-1})$, the geometric inertia elements, written in $G = \mathbb{Z}/A\mathbb{Z} \times \mathbb{Z}/B\mathbb{Z} \times \mathbb{Z}/B\mathbb{Z}$, are

$$v_1 = (-1, -A, -1), \qquad v_2 = (-1, -A, -t).$$

Their difference is $(0, 0, t-1)$, which generates the third factor because $\gcd(t-1, B) = 1$. Combining a suitable multiple of this element with v_1 gives $(-1, -A, 0)$. Its order is AB: the first coordinate has order A, the second has order B since $\gcd(A, B) = 1$. Thus v_1, v_2 generate all of G. Equivalently, no nontrivial character annihilates both inertia elements. The subgroup generated by geometric inertia is therefore the full deck group, proving geometric connectedness of the product cover and of every character quotient.

All covering degrees are odd, so wild inertia is trivial. For a rank-one character its local Artin conductor is one if the inertia character is nontrivial, and zero otherwise. Thus its conductor degree is the number r of active rows of (26), and $r \le 8$. Cancellation can occur at individual rows; we do not presume ramification at every boundary point. For a nontrivial character triple the inertia test proves geometric nontriviality and $r > 0$. The genus-one Artin-character Weil bound now gives $(2 \cdot 1 - 2 + r)\sqrt{t} \le 8\sqrt{t}$; see [10, §4, Lemma 1] and the Lang construction in [9, pp. 198–199]. In cohomological terms, extend over the inactive boundary points. On the complement of the r active points, H_c^0, H_c^2 vanish and the tame Euler-characteristic formula gives $\dim H_c^1 = 2g - 2 + r = r$.

The eigenvalues have absolute value at most $\sqrt{t}$, by the Weil theorem on the character component of the finite cover. No active boundary point is K-rational, so this trace sum is exactly the sum over $X_b(K)$, with no rational boundary correction. □

Corollary 5.22 (The precise two-character exception)**.** *Let $b \in U$ satisfy $b^3 \neq 1$ and let $\xi, \zeta \in \widehat{C_d}$. Use the character decomposition lemma and* (21)*. If $(\xi, \zeta) \in \mathcal{E}_{\mathrm{norm}}$, then*

$$S_b(\xi, \zeta) = \alpha(b) \# X_b(K).$$

In every other case $|S_b(\xi, \zeta)| \leq 8\sqrt{t}$. In particular, the assertion for every nontrivial two-character pair is false: a nontrivial $\alpha \circ N_{L/E}$ with $\zeta = 1$ is constant on the norm fiber.

Proof. The decomposition lemma identifies the exception set with the trivial varying triple. That triple gives the exact constant sum. Every pair outside the set has a nontrivial varying triple, so Proposition 5.21 gives the bound. □

Lemma (Local averaging in a cyclic group)**.** *Put*

$$\theta_2(n) = \prod_{p \mid n} (1 - p^{-2}), \qquad J_b = \{p \mid A : b \in U^p\},$$
$$\lambda_A(b) = \prod_{p \in J_b} (1 - p^{-1}), \quad H(A) = \prod_{p \mid A} \frac{p+2}{p+1}.$$

The following are exact averages over all A elements of U:

$$\begin{aligned} \frac{1}{A} \sum_{b \in U} \lambda_A(b) &= \theta_2(A), \\ \frac{1}{A} \sum_{b \in U} \lambda_A(b) 2^{|J_b|} &= \theta_2(A) H(A). \end{aligned} \tag{27}$$

In particular the average of $\lambda_A(b)\theta_2(B)$ is $\theta_2(A)\theta_2(B) = \theta_2(d)$.

Proof. Fix a generator u and write $b = u^j$, $j \in \mathbb{Z}/A\mathbb{Z}$. For $p \mid A$ the condition $b \in U^p$ is equivalent to $p \mid j$. Reduction of uniform j modulo $\mathrm{rad}(A)$ is uniform; the Chinese remainder theorem makes the residues modulo the distinct primes independent. The probability of $p \in J_b$ is therefore $1/p$. The local expectation for λ_A is

$$(1 - 1/p) + \frac{1}{p}(1 - 1/p) = 1 - p^{-2}.$$

For $\lambda_A(b) 2^{|J_b|}$ it is

$$(1 - 1/p) + \frac{2}{p}(1 - 1/p) = (1 - p^{-2}) \frac{p+2}{p+1}.$$

Multiplication of these independent local expectations proves both identities. Since $\gcd(A, B) = 1$, their prime supports are disjoint, which proves the last equality. □

Theorem 5.23 (Joint generation density at the inner boundary)**.** *Use $\theta_2, \lambda_A, J_b, H(A)$ from the local averaging lemma, and put $W(n) = 2^{\omega(n)}$, $r_0 = \gcd(3, A) \leq 3$. Let $\mathcal{R}_{d,b} = \mathcal{R}_d \cap X_b(K)$. For every $b \in U$ with $b^3 \neq 1$,*

$$\big| |\mathcal{R}_{d,b}| - \lambda_A(b)\theta_2(B)A \big| \leq \lambda_A(b)\theta_2(B)\big(8\, 2^{|J_b|} W(B) - 6\big)\sqrt{t}. \tag{28}$$

For the full family,

$$\big||\mathcal{R}_d| - \theta_2(d)A^2\big| \le \theta_2(d)A\big(8H(A)W(B) - 6\big)\sqrt{t} + 2r_0A. \tag{29}$$

Since $H(A) \le W(A)$, this implies

$$|\mathcal{R}_d| = \theta_2(d)t^2 + O(t^{3/2}W(d)), \qquad |\mathcal{R}_d| \sim \theta_2(d)t^2.$$

The fixed-fiber population is $O(t)$, and its coefficient depends on b. If b is primitive in U, then J_b is empty and the coefficient is $\theta_2(B)$. Moreover $|T_d - A^2| \le 2A\sqrt{t} + r_0A$, so the actual proportion of jointly generating slopes among the ordinary intersection is $|\mathcal{R}_d|/T_d = \theta_2(d) + O(W(d)/\sqrt{t})$.

Proof. The group isomorphism gives

$$\langle a, 1+a\rangle = C_d \quad \Longleftrightarrow \quad \langle b, w\rangle = U \text{ and } \langle v, v_+\rangle = V.$$

For $p \mid A$ with $b \notin U^p$ the first condition is automatic. For $p \in J_b$ it requires $w \notin U^p$. By orthogonality its indicator is the product of the factors

$$1 - \frac{1}{p}\sum_{\eta^p=1}\eta(w), \qquad p \in J_b.$$

Each factor has constant coefficient $1 - 1/p$ and total absolute coefficient sum $2(1 - 1/p)$. The second condition has indicator

$$\prod_{p\mid B}\left(1 - \frac{1}{p^2}\sum_{\chi^p=\psi^p=1}\chi(v)\psi(v_+)\right).$$

Here each factor has constant coefficient $1 - 1/p^2$ and total absolute coefficient sum $2(1 - 1/p^2)$. Characters belonging to different prime-primary components have unique products, so multiplying these expansions gives constant coefficient $c_b = \lambda_A(b)\theta_2(B)$ and total absolute coefficient sum $c_b 2^{|J_b|}W(B)$. The Hasse error for the trivial triple is $2c_b\sqrt{t}$; Proposition 5.21 bounds all other terms by $8\sqrt{t}$. This proves (28).

The local averaging lemma supplies (27). Multiplication by $\theta_2(B)$ turns $\theta_2(A)$ into $\theta_2(d)$; it also turns the weighted average into $\theta_2(d)H(A)$. These identities average over all $b \in U$, before separating the exceptional fibers.

There are precisely r_0 exceptional b with $b^3 = 1$. For every b, each $w \in U$ allows at most two roots of $X^2 + (1 + b + w)X + b$. Thus $\#X_b(K) \le 2A$ even in an exceptional fiber. Its contribution differs from the formal main term $c_bA \in [0, A]$ by at most $2A$. Sum (28) over the good fibers, bound the positive error coefficients by their sum over all b, and use the two exact averages. This gives (29). It includes all exceptional and subfield points, rather than applying the smooth-fiber bound to them. Finally $W(d) = d^{o(1)} = t^{o(1)}$, while $\theta_2(d) \ge 1/2$: the latter follows by comparison with the telescoping product $\prod_{n=2}^{\infty}(1 - n^{-2}) = 1/2$. Replacing $A^2 = (t+1)^2$ by t^2 gives the stated asymptotic. For T_d sum the Hasse bound over the good fibers and use $|\#X_b(K) - A| \le A$ on each exceptional fiber. This proves its bound and, by division, the density assertion. □

Theorem 5.24 (Double full-order statistics and all-dimensional existence)**.** *Define $\theta(n) = \varphi(n)/n$ and*

$$D_{d,b} = \#\{a \in X_b(K) : \operatorname{ord}(a) = \operatorname{ord}(1+a) = d\}, \qquad D_d = \sum_{b\in U} D_{d,b}.$$

If b is not primitive in U, then $D_{d,b} = 0$. For primitive b, put $c = \theta(A)\theta(B)^2$. Then

$$|D_{d,b} - cA| \le c\big(8W(A)W(B)^2 - 6\big)\sqrt{t}. \tag{30}$$

In particular,

$$|D_d - \theta(d)^2A^2| \le \theta(d)^2A\big(8W(A)W(B)^2 - 6\big)\sqrt{t}, \tag{31}$$

and $D_d \sim \theta(d)^2t^2$ as $s \to \infty$. Define the explicit analytic range

$$\mathcal{S}_{\mathrm{an}} = \{s \in \mathbb{Z}_{\ge 2} : A > (8W(A)W(B)^2 - 6)\sqrt{t}\}, \tag{32}$$

with $t = 2^s$, $A = t + 1$, and $B = t^2 + 1$. For every $s \in \mathcal{S}_{\mathrm{an}}$ and every primitive $b \in U$, $D_{d,b} > 0$. This set contains all sufficiently large integers s. Moreover, $D_d > 0$ for every integer $s \ge 2$.

Proof. Order d forces $a^B = b$ to have order A, which proves the vanishing assertion. For primitive b, the two full-order conditions are exactly that w is primitive in U and v, v_+ are primitive in V. Apply the primitive-character indicator proved in Theorem 5.19 separately to these three variables. For a cyclic group of order n it has constant coefficient $\theta(n)$ and total absolute coefficient sum $\theta(n)W(n)$. The product therefore has constant coefficient c and total absolute coefficient sum $cW(A)W(B)^2$. The trivial triple contributes $c\#X_b(K)$, with Hasse error $2c\sqrt{t}$; every other triple has the mixed bound $8\sqrt{t}$. This proves (30). Sum over the $\varphi(A) = A\theta(A)$ primitive norm fibers and use $\theta(d) = \theta(A)\theta(B)$ to obtain (31). The relative error tends to zero because $W(A)W(B)^2 = t^{o(1)}$. The stated inequality makes the lower bound positive in every primitive norm fiber. For all-dimensional existence, Proposition 5.25 below gives $D_d \ge \Delta_t > 0$ for $s \ge 33$, independently of the preceding three-character sieve. The exact certificates in Appendix A prove $D_d > 0$ for each $2 \le s \le 32$. □

Proposition 5.25 (A trace-one double full-order subfamily)**.** *For $t = 2^s$, $s \ge 2$, use $A = t + 1$, $B = t^2 + 1$ and $d = AB$ as above. Put*

$$\Delta_t = \#\{\gamma \in V : \tau = \gamma + \gamma^{-1} \in U,\ \mathrm{ord}(\gamma) = B,\ \mathrm{ord}(\tau) = A\}.$$

Every element counted by Δ_t gives, injectively,

$$a = (1 + \gamma)^{-1} \in C_d, \qquad 1 + a = a^{t^2}, \qquad \mathrm{ord}(a) = \mathrm{ord}(1 + a) = d. \tag{33}$$

In particular $D_d \ge \Delta_t$. For any characters η of U and χ of V with $(\eta, \chi) \ne (1, 1)$,

$$\left| \sum_{\substack{\gamma \in V \\ \gamma + \gamma^{-1} \in U}} \eta(\gamma + \gamma^{-1})\chi(\gamma) \right| \le 4\sqrt{t}. \tag{34}$$

Consequently

$$|\Delta_t - \theta(d)(t + 1)| \le \theta(d)\big(4W(d) - 2\big)\sqrt{t}. \tag{35}$$

The sufficient condition $\sqrt{t} \ge 4W(d)$ implies $\Delta_t > 0$; it holds for every $s \ge 33$ by the integer bounds in Appendix A.

Proof. The full-order construction.

Here $\gamma^{t^2} = \gamma^{-1}$ and $\gamma \ne 1$. Thus $a^{t^2} = \gamma/(1 + \gamma) = 1 + a$. Also

$$N_{L/E}(1 + \gamma) = \gamma + \gamma^{-1} = \tau, \qquad a^B = \tau^{-1}, \qquad a^{t^2-1} = \gamma.$$

Since $N_{E/K}(\tau) = 1$, the norm of a to K is one, so $\mathrm{ord}(a)$ divides d. The two displayed powers force both A and B to divide this order. Their gcd is one, and hence the order is d. The conjugate $1 + a$ has the same order. The formula for a is injective in γ.

The curve and its four boundary points.

We analyze the norm fiber X_1 directly, rather than applying Proposition 5.21, whose good-fiber hypothesis excludes this parameter. Write its variable as γ. Over $\overline{K}$, the relative norm-one torus has coordinates (x, y, x^{-1}, y^{-1}). Its K-points have $x = \gamma$, $y = \gamma^t$, $\gamma \in V$, and its Frobenius is

$$F_Y(x, y) = (y^{-t}, x^t).$$

The condition $N_{E/K}(\gamma + \gamma^{-1}) = 1$ gives the projective closure $Y_{\overline{K}} \subset \mathbb{P}^1 \times \mathbb{P}^1$ of

$$(x + 1)^2(y + 1)^2 = xy. \tag{36}$$

Here Y denotes the K-curve with the inherited torus descent F_Y, not the split curve equipped with coordinatewise t-power Frobenius. The descent extends to the projective closure and preserves its boundary. In particular the relation $y = x^t$ is imposed on K-points, not in its geometric function field. On the affine chart the two partial derivatives are y and x; their only common zero $(0, 0)$ is not on the curve. The boundary is

$$P_0 = (0, 1), \quad P_\infty = (\infty, 1), \quad Q_0 = (1, 0), \quad Q_\infty = (1, \infty).$$

At P_0, with $u = y+1$, the equation is $(1+x)^2u^2+x(1+u) = 0$, whose derivative with respect to x is one at the point. Thus it is smooth and $v_{P_0}(x) = 2$. Reciprocating x or y, and interchanging them, gives the same smoothness test at the other boundary points. This proves that the bidegree-$(2, 2)$ curve is smooth. It is geometrically irreducible. Smoothness implies that the curve is reduced and that distinct irreducible components would be disjoint. Every component has a nonzero effective bidegree (a_i, b_i), with $a_i, b_i \geq 0$ and $\sum_i (a_i, b_i) = (2, 2)$. If one component has both entries positive, its intersection $a_ib_j + a_jb_i$ with every other component is positive. If no component has both entries positive, the sum $(2, 2)$ forces a pure vertical component and a pure horizontal component, whose intersection is again positive. Both possibilities contradict disjointness. Adjunction on $\mathbb{P}^1 \times \mathbb{P}^1$ now gives genus $(2 - 1)(2 - 1) = 1$. Frobenius permutes the four boundary points in the cycle

$$P_0 \longmapsto Q_0 \longmapsto P_\infty \longmapsto Q_\infty \longmapsto P_0.$$

There is no K-rational boundary point. Therefore the open curve Y° has exactly the points in the sum (34), and $|\#Y^\circ(K) - (t + 1)| \leq 2\sqrt{t}$ by Hasse–Weil.

Actual Lang covers and their character traces.

Let $\mathcal{U} = \ker N_{E/K}$ and $\mathcal{V} = \mathrm{Res}_{E/K}(\ker N_{L/E})$; their K-point groups are U and V. On Y° use the maps $\tau = \gamma + \gamma^{-1}$ to $\mathcal{U}$ and γ to $\mathcal{V}$ itself. Write

$$w_0 = (x + 1)^2/x, \qquad h = y^t/x.$$

The split coordinates of τ are (w_0, w_0^{-1}). The two Lang pullbacks, with their inherited K-structures, are

$$z_U^A = w_0^{-1}, \qquad z_V^B = h. \tag{37}$$

For the first, the Lang equation is $z_U^{-t}/z_U = w_0$. For the second, the split equations are $z_1^{-t}/z_0 = x$, $z_0^t/z_1 = y$; elimination gives $z_0^B = y^t/x$, with $z_V = z_0$. Thus the actual functions are w_0^{-1} and h,

not merely a geometrically convenient Kummer model. Above F_Y, Frobenius on the two covering coordinates is

$$(z_U, z_V) \longmapsto (w_0 z_U, x z_V).$$

This follows from the split Lang equations. It also respects (37), since $w_0(F_Y) = w_0^{-t}$ and $h(F_Y) = x^B h$. At a K-point the deck multipliers are exactly τ and γ. Choosing the character convention for these multipliers gives trace $\eta(\tau)\chi(\gamma)$; the inverse convention just inverts both characters.

Geometric connectedness and conductor.

The complete boundary valuation table is

	w_0	h
P_0	-2	-2
P_∞	-2	2
Q_0	2	$2t$
Q_∞	2	$-2t$

(38)

For example $x+1$ has simple zeros at Q_0, Q_∞ and a double pole at P_∞; y has a double zero at Q_0 and a double pole at Q_∞. These observations give every entry of the table. Suppose for $0 \le e_0 < A$, $0 \le e_1 < B$ that

$$(w_0^{-1})^{Be_0} h^{Ae_1} \in \overline{K}(Y)^{\times AB}.$$

Its valuation at P_0 is $2Be_0 - 2Ae_1$. Reduction modulo A and modulo B, using that A, B are odd and coprime, gives $A \mid e_0$ and $B \mid e_1$. This single valuation separates the two cyclic components because reduction modulo each order kills the other component. Thus $e_0 = e_1 = 0$. The Kummer criterion proves that the product cover in (37) is geometrically connected with full group $U \times V$; in particular every nontrivial character quotient is geometrically nontrivial.

All cover degrees are odd, so the ramification is tame. At every boundary point the valuation of w_0^{-1} is one of ± 2, and that of h is one of ± 2 or $\pm 2t$. Since A and B are odd, $\gcd(2, A) = \gcd(2t, B) = 1$; each coordinate therefore generates its cyclic deck factor. A nontrivial character on either factor is nontrivial on that coordinate, and the two character values cannot cancel because their orders divide the coprime integers A and B. Thus every nontrivial pair is ramified at each of the four boundary points, with Swan conductor zero and local Artin conductor one. Its total conductor is exactly four, and it is unramified on Y°. The associated rank-one finite-monodromy sheaf is pure of weight zero and geometrically nontrivial. Thus $H_c^0 = H_c^2 = 0$, and Grothendieck–Ogg–Shafarevich gives $\dim H_c^1 = 2g(Y) - 2 + 4 = 4$. The classical curve character bound [10, Lemma 1], with the Lang construction as in [9], bounds each Frobenius eigenvalue by $\sqrt{t}$. This proves (34).

The two primitive indicators and positivity.

Apply the primitive-character indicator to $\tau \in U$ and $\gamma \in V$. Their product has constant coefficient $\theta(A)\theta(B) = \theta(d)$ and total absolute coefficient sum $\theta(d)W(A)W(B) = \theta(d)W(d)$. The trivial pair has Hasse error $2\theta(d)\sqrt{t}$. The other pairs have total coefficient mass $\theta(d)(W(d) - 1)$ and character bound $4\sqrt{t}$. Their sum is exactly the error in (35). If $\sqrt{t} \ge 4W(d)$, then $(t+1) - (4W(d) - 2)\sqrt{t} > 0$, proving positivity. For $33 \le s \le 113$, the complete integer certificates in Appendix A give $\omega(d) \le r_s$ with $s \ge 4 + 2r_s$, hence this condition. For $s \ge 114$, the same appendix proves for every odd n that $W(n) < 4096 n^{1/8}$. Since $d < 2t^3$,

$$(4W(d))^8 < 2^{113} t^3 < t^4 = (\sqrt{t})^8.$$

These two ranges prove $\Delta_t > 0$ for every $s \ge 33$. □

The construction (33) is the classical trace-one reciprocal-trace change of variable [20, §4], applied here with the additional condition $N_{E/K}(\tau) = 1$ and the smaller exact order $A = t + 1$. The estimate established above concerns this specified fiber; the Frobenius identity itself is not a priority claim. It also implies $\mathrm{Tr}_{L/K}(a) = (a + a^{t^2}) + (a^t + a^{t^3}) = 1 + 1 = 0$, so these particular slopes are not normal over K.

The three populations $|\mathcal{R}_d|$, N_d^{prim} and D_d have different main terms. Theorem 5.24 now gives $D_d > 0$ for every $s \geq 2$, using the trace-one subfamily of Proposition 5.25 for $s \geq 33$ and the exact double-order certificates for $2 \leq s \leq 32$. This subfamily has a linear main term $\theta(d)(t+1)$; it does not replace the quadratic full-family main term $\theta(d)^2(t+1)^2$. Nor does all-dimensional existence assert positivity in every primitive norm fiber at small s. Indeed $\Delta_t = 0$ at $s = 3, 5$; the finite certificates at these parameters use slopes outside this trace-one subfamily.

Theorem 5.26 (The six-dimensional Singer application)**.** $\langle T, D \rangle = \mathrm{AGL}(1, 64)$*; its order is* $4032 = 64 \cdot 63$.

Proof. By Proposition 3.3, in field coordinates the linear part of T is multiplication by a primitive $\alpha \in \mathbb{F}_{64}$. The translation is $c = T(0) = 63 \neq 0$. Thus $U = V$ and the least common multiple in Corollary 5.2 is already 63 because $\mathrm{ord}(\alpha) = 63$. The corollary proves the claim. □

The general structure allows reducible and nonsemisimple matrices. For example, the four-dimensional finite check gives the following distribution:

group order	3	12	15	48	192	240
number of maps	112	3360	4032	5040	20160	60480

Of the 93184 affine complete mappings, 80640 give transitive groups; 60480 give primitive groups, all doubly transitive in this dimension. These counts are computational evaluations of the structure theorem, reproduced by the additional verifier in Appendix A; the action criteria themselves are proved above for every dimension.

6 Uniform statement for $2k$-bit words

Identify the $2k$-dimensional $\mathbb{F}_2$-space V_{2k} with $\mathbb{F}_{2^{2k}}$; write $M = 2^{2k} - 1 = (2^k - 1)(2^k + 1)$. In field coordinates an affine Singer layer is a map $T(x) = \alpha x \oplus b$ whose linear part is multiplication by a primitive element α (order M).

Lemma 6.1. *For every* $k \geq 1$, $\mathrm{ord}_{2^k+1}(2) = 2k$.

Proof. Let $m = \mathrm{ord}_{2^k+1}(2)$. Since $2^k \equiv -1 \pmod{2^k + 1}$, we have $m \mid 2k$, while $m \nmid k$. Suppose, for contradiction, that $m < 2k$. If m were odd, $m \mid 2k$ would imply $m \mid k$. Thus m is even; write $m = 2d$ with $d \mid k$, and $m < 2k$ gives $d < k$. The subgroup $\langle 2 \rangle \subset (\mathbb{Z}/(2^k + 1)\mathbb{Z})^\times$ is cyclic of order $2d$, so its unique element of order 2 is 2^d. But $2^k \equiv -1$ also has order 2 in this subgroup; hence $2^d \equiv -1 \pmod{2^k + 1}$. Thus $2^k + 1 \mid 2^d + 1$, contradicting $0 < 2^d + 1 < 2^k + 1$. Therefore $m = 2k$. □

For $q = 2^k$, define

$$\mathrm{PTE}(k): \quad \exists \beta \in \mathbb{F}_{q^2}^*, \quad \mathrm{ord}(\beta) = q^2 - 1, \quad \mathrm{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(\beta) = 1.$$

The following theorem supplies the required existence input; its proof is given in Appendix B.

Theorem 6.2 (Classical PTE existence)**.** *For every* $k \geq 1$, $\mathrm{PTE}(k)$ *holds.*

Proof. See Appendix B. □

Theorem 6.3. *For every $k \geq 1$ there exists a primitive $\alpha \in \mathbb{F}_{2^{2k}}$ such that, for every $b \in \mathbb{F}_{2^{2k}}^*$, the affine Singer layer $T_b(x) = \alpha x \oplus b$ has $D_b = (\mathrm{id} \oplus T_b)^{-1}$ and $K_b = (\mathrm{id} \oplus D_b)^{-1}$ defined. The maps T_b, D_b have cycle type $1 + M$, while K_b has cycle type $1 + (2^k - 1) \times (2^k + 1)$, and $\langle T_b, D_b \rangle = \mathrm{AGL}(1, 2^{2k})$. The triality operator satisfies $F^3 = \mathrm{id}$ with $T_b \mapsto D_b \mapsto K_b \mapsto T_b$.*

Proof. Let $b \neq 0$ be arbitrary and write $T = T_b$, $D = D_b$, $K = K_b$. The linear parts of T, D, K are α, $(1 \oplus \alpha)^{-1}$, and $\kappa = 1 \oplus \alpha^{-1} = (1 \oplus \alpha)/\alpha$ respectively (independent of the translation b); the three slopes are cycled by the map $z \mapsto 1/(z \oplus 1)$, so $F^3 = \mathrm{id}$. Since $\mathbb{F}_{2^{2k}}^*$ is cyclic, an affine map with slope $\mu \neq 1$ has cycle type $1 + \frac{M}{\mathrm{ord}(\mu)} \times \mathrm{ord}(\mu)$ (the translation only moves the fixed point; a shorter orbit would force T^d, $d < \mathrm{ord}(\mu)$, to have a second fixed point, impossible since $\mu^d \neq 1$); the stated types therefore follow once $\mathrm{ord}(\alpha) = \mathrm{ord}(1 \oplus \alpha) = M$ and $\mathrm{ord}(\kappa) = 2^k + 1$. Existence of such an α follows directly from Theorem 6.2. Put $q = 2^k$ and choose a primitive $\beta \in \mathbb{F}_{q^2}$ with $\beta^q \oplus \beta = 1$. Take $\alpha = \beta$. Then $1 \oplus \alpha = \alpha^q$, so

$$\mathrm{ord}(\alpha) = \mathrm{ord}(1 \oplus \alpha) = q^2 - 1 = M, \qquad \kappa = \frac{1 \oplus \alpha}{\alpha} = \alpha^{q-1}, \qquad \mathrm{ord}(\kappa) = q + 1.$$

Here the last equality follows from $\mathrm{ord}(\alpha^{q-1}) = M/\gcd(M, q-1) = q + 1$. These are exactly the required slope orders. To make the dependence on b explicit, set $S_b(x) = bx$. Then $T_b = S_b T_1 S_b^{-1}$. Indeed, for any invertible additive map S and $g = SfS^{-1}$,

$$(\mathrm{id} \oplus g)(x) = S\big((\mathrm{id} \oplus f)(S^{-1}x)\big).$$

Taking inverses gives $F(g) = SF(f)S^{-1}$ whenever $F(f)$ is defined. Applying this with $S = S_b$ gives $D_b = S_b D_1 S_b^{-1}$ and $K_b = S_b K_1 S_b^{-1}$. Their fixed points are

$$\mathrm{Fix}(T_b) = \{b/(1 \oplus \alpha)\}, \qquad \mathrm{Fix}(D_b) = \{b/\alpha\}, \qquad \mathrm{Fix}(K_b) = \{b\}.$$

Thus all nonzero b give the stated cycle types; §5 gives the group-generation claim. □

Since the K-slope has order $2^k + 1$, the degree over $\mathbb{F}_2$ of its minimal polynomial is $\mathrm{ord}_{2^k+1}(2) = 2k$ by Lemma 6.1; this polynomial is therefore an irreducible degree-$2k$ factor of Φ_{2^k+1}. **Remark (the τ-collapse).** Put $q = 2^k$. For $\gamma \in \mathbb{F}_{q^2}^*$ of order $q + 1$, one has $\gamma^q = \gamma^{-1}$; hence $\tau = \gamma \oplus \gamma^{-1}$ satisfies $\tau^q = \tau$ and belongs to $\mathbb{F}_q$. The Frobenius identities $(1 \oplus \gamma)^{2^k-1} = \gamma^{-1}$ and $(1 \oplus \gamma)^{2^k+1} = \tau$, together with $\gcd(2^k - 1, 2^k + 1) = 1$, give the following order calculation. For an element z of finite order N and coprime positive integers r, s, $\mathrm{ord}(z^r) = N/\gcd(N, r)$ implies $N = \mathrm{lcm}(\mathrm{ord}(z^r), \mathrm{ord}(z^s))$. Consequently $\mathrm{ord}(1 \oplus \gamma) = \mathrm{lcm}(\mathrm{ord}(\gamma^{-1}), \mathrm{ord}(\tau)) = \mathrm{lcm}(2^k + 1, 2^k - 1) = M$ whenever τ is primitive in $\mathbb{F}_{2^k}$; the pair $(u^{2^k-1}, u^{2^k+1}) = (\gamma^{-1}, \tau^{-1})$ for $u = \gamma/(1 \oplus \gamma)$ gives the same order for u. Since $\alpha = (1 \oplus \gamma)^{-1}$ and $\kappa = 1 \oplus \alpha^{-1} = \gamma$, once γ has order $2^k + 1$ the three order conditions on α reduce to the single condition that τ be primitive. The order assumption on γ is essential; it is not implied by τ being primitive alone. Existence of such a γ is the classical square-order statement (see [21] and [22, Theorem 1, p. 1]; for an equivalent square-order formulation, see [20, Theorem 1.5, p. 170]); it is also supplied directly by Theorem 6.2. For $q = 2^k$ the appendix proof is self-contained, so the paper does not use these classical results as external existence inputs.

The theorem is an existence statement in admissible coordinates (equivalently, for an admissible ordered generator basis). The argument of §5 applies verbatim with $Q = 2^{2k}$ (taking $b \neq 0$ in the proof), so the generation $\langle T, D \rangle = \mathrm{AGL}(1, 2^{2k})$ holds for every k. For the concrete coordinate presentation of §3 the orders and cycle data were verified for $k = 1, 2, 3, 4$ (the bases are listed in the appendix): $k = 1$ (2 bits) order $12 = 4 \cdot 3$, K type $1 + 1 \cdot 3$, factor Φ_3 (basis $(1, 3)$: $\mathrm{Fix}(T) = \{1\}$, $\mathrm{Fix}(D) = \{2\}$, $\mathrm{Fix}(K) = \{3\}$); $k = 2$ (4 bits) order $240 = 16 \cdot 15$, K type $1 + 3 \cdot 5$, factor Φ_5;

$k = 3$ (6 bits) order $4032 = 64 \cdot 63$, K type $1 + 7 \cdot 9$, factor Φ_9 (this paper); $k = 4$ (8 bits) order $65280 = 256 \cdot 255$, K type $1 + 15 \cdot 17$, linear part an irreducible degree-8 factor of Φ_{17}.

The basis is not canonical even in dimension 6: among the 720 orderings of these six generators, 572 do not even make $\mathrm{id} \oplus T$ a permutation; of the 148 that do, only 54 give T of order 63, and only 9 give the exact cycle-type triple $T = D = 1 + 63$, $K = 1 + 7 \cdot 9$. For instance $(4\,6)$ keeps $T = 1 + 63$ but changes $\mathrm{Fix}(T)$ from 38 to 11 and the cycle types of D and K to $D = 1 + 3 \cdot 21$, $K = 1 + 63$. Hence the particular entries of G and the anchors $\{38, 41, 63\}$ are coordinates of one representative, not invariants, and no uniqueness statement is made. The structural results of §5–6, however, are representative-independent: $\langle T, D \rangle = \mathrm{AGL}(1, 64)$, the third-iterate identity $F^3 = \mathrm{id}$, and the $2k$-bit existence statement hold for any admissible basis; only the numerical anchors change with the basis.

Remark (constructiveness).
Theorem 6.3 is an existence statement; its existence input is proved in Appendix B. The τ-collapse turns it into a finite search with certification. For $\tau = \gamma \oplus \gamma^{-1} \in \mathbb{F}_{2^k}$, the Artin–Schreier identity ($x^2 + \tau x + 1$ *is irreducible over* $\mathbb{F}_{2^k}$ *iff* $\mathrm{Tr}_{\mathbb{F}_{2^k}/\mathbb{F}_2}(1/\tau) = 1$) gives 2^{k-1} candidate values of τ. For a candidate, fix an explicit representation of $\mathbb{F}_{2^{2k}}$ and take a root γ of $x^2 + \tau x + 1$ (found, for instance, by exhaustive search); the collapse gives the required orders for $1 \oplus \gamma$ and $\gamma/(1 \oplus \gamma)$ precisely when τ is primitive *and* $\mathrm{ord}(\gamma) = 2^k + 1$. The second condition is not automatic from the first: for example, when $k = 6$ and $\mathbb{F}_{64} = \mathbb{F}_2[t]/(t^6 + t + 1)$, the element $\tau = t^5 + t^3 + 1$ is primitive and $\mathrm{Tr}(1/\tau) = 1$, but the roots of $x^2 + \tau x + 1$ have order 13, not 65. The theorem of Appendix B guarantees that a jointly successful candidate exists, so a deterministic enumeration that tests both conditions terminates. Certifying them requires factoring $2^k - 1$ and $2^k + 1$ (or an equivalent exact-order test). This procedure supplies a deterministic construction with certification; no factorization-free closed formula is supplied here. A necessary restatement is that $u = 1/\tau$ is a primitive element of $\mathbb{F}_{2^k}$ with $\mathrm{Tr}_{\mathbb{F}_{2^k}/\mathbb{F}_2}(u) = 1$; equivalently, its degree-k minimal polynomial has x^{k-1}-coefficient 1. This condition alone is still not sufficient for admissibility, because the root-order condition above must also be checked.

7 Fixed mappings and orbits under F

Lemma 4.2 implies that every orbit under F has length one or three. We first give the scalar classification, then describe all fixed complete mappings geometrically and count the orbits in the entire affine class. An *F-fixed map* means $F(f) = f$; this is distinct from the unique fixed point of the permutation f in Lemma 4.1.

Scalar affine maps

Proposition 7.1. *Let $k \geq 1$, $Q = 2^{2k}$, and $\mathcal{A}_Q = \{f_{a,c} : x \mapsto ax \oplus c \mid a \in \mathbb{F}_Q \setminus \{0, 1\},\ c \in \mathbb{F}_Q\}$. The operator F acts on $\mathcal{A}_Q$ with two singleton orbits and $\big(Q(Q-2) - 2\big)/3$ orbits of length three. Its fixed maps are exactly $f_{a,0}$ with $a^2 + a + 1 = 0$. On the invariant subset $c \neq 0$, every orbit has length three.*

Proof. In coefficient pairs,

$$F(a, c) = \left((1 + a)^{-1}, \frac{c}{1 + a} \right).$$

The new slope again lies outside $\{0, 1\}$, and Lemma 4.2 gives $F^3 = \mathrm{id}$, so each orbit has length one or three. A fixed pair must satisfy $a = (1 + a)^{-1}$, equivalently $a^2 + a + 1 = 0$; the constant-term equation is $c(1 + a) = c$, so $ac = 0$ and $c = 0$. Conversely these conditions give a fixed pair. The

polynomial $a^2 + a + 1$ has two distinct roots in $\mathbb{F}_Q$, since $\mathbb{F}_4 \subseteq \mathbb{F}_Q$. There are $Q(Q-2)$ pairs in total, yielding the stated counts. Multiplication of c by $(1+a)^{-1} \neq 0$ preserves $c \neq 0$, and no fixed pair lies in that subset. □

Fixed maps and line spreads

A *line spread* of $V = \mathbb{F}_2^n$ is a collection of two-dimensional $\mathbb{F}_2$-subspaces whose nonzero vectors partition $V \setminus \{0\}$. Thus a line here is two-dimensional as a vector space; it is a projective line in $\mathrm{PG}(n-1, 2)$. Each such subspace has exactly three nonzero vectors $x, y, x \oplus y$. A cyclic orientation on a line specifies one of the two 3-cycles on them.

Proposition 7.2. *For $n \geq 1$, the F-fixed complete mappings of $\mathbb{F}_2^n$ are in bijection with line spreads equipped with a cyclic orientation on each line. Every such map fixes 0 and has only 3-cycles on the nonzero vectors. They exist exactly when n is even. If s_n denotes the number of line spreads in a fixed coordinate space, their number, for even n, is*

$$s_n \, 2^{(2^n-1)/3}.$$

Nonlinear F-fixed complete mappings exist for every even $n \geq 4$.

Proof. If $F(f) = f$, then $f^{-1} = \mathrm{id} \oplus f$. Evaluating at $f(x)$ gives

$$f^2(x) = x \oplus f(x) = f^{-1}(x), \qquad f^3 = \mathrm{id}.$$

Let p be the unique fixed point from Lemma 4.1. The displayed equality at p gives $p = p \oplus p = 0$. Thus each nonzero cycle is $(x, f(x), x \oplus f(x))$, the nonzero part of a two-dimensional subspace. Disjoint cycles supply a line spread and their orientations. Conversely, orient each line of a spread and fix 0. The resulting permutation satisfies $f^2 = \mathrm{id} \oplus f = f^{-1}$, so it is complete and $F(f) = f$. This proves the bijection and the count.

A spread requires $3 \mid 2^n - 1$, hence even n. If $n = 2m$, identify V with $\mathbb{F}_4^m$; its one-dimensional $\mathbb{F}_4$-subspaces give a line spread over $\mathbb{F}_2$, proving existence. For $m \geq 2$ (equivalently, even $n \geq 4$), start with the linear map $A : x \mapsto \omega x$, where $\omega^2 + \omega + 1 = 0$ in $\mathbb{F}_4$, and reverse the orientation on exactly one spread line U. The resulting map f equals A^2 on $U \setminus \{0\}$ and equals A elsewhere, and remains F-fixed. For $0 \neq u \in U$ and $v \notin U$, one has $u \oplus v \notin U$, and

$$f(u \oplus v) = Au \oplus Av \neq A^2 u \oplus Av = f(u) \oplus f(v),$$

since $(A^2 + A)u = u \neq 0$. Hence f is nonlinear. □

If V is identified with $\mathbb{F}_{2^n}$ for even n, the spread obtained from $\mathbb{F}_4$ consists of the multiplicative cosets of $\mathbb{F}_4^*$. An orientation on each coset is multiplication by either ω or ω^2. Thus these examples are a particular family of the first-order cyclotomic complete mappings considered in [25]. Proposition 7.2 describes fixed maps from arbitrary line spreads, without choosing a field structure or restricting to this coset partition.

For $n = 4$, exact-cover enumeration of the 35 lines on 15 nonzero vectors gives $s_4 = 56$, recovering the classical spread count also formalized in Coq by Magaud [3]. The proposition then gives $56 \cdot 2^5 = 1792$ fixed complete mappings. Enumeration of their orientations finds 112 linear and 1680 nonlinear maps, in agreement with the affine count below. These finite counts, the complete list of spreads, and a nonlinear witness are reproduced by `verify_complete_mapping_orbits.py`.

All affine complete mappings

Write $h_j = |\mathrm{GL}(j,2)| = \prod_{i=0}^{j-1}(2^j - 2^i)$, where $h_0 = 1$ is the empty product, and let

$$\mathcal{C}_n^{\mathrm{aff}} = \{x \mapsto Lx \oplus c : \ L, I+L \in \mathrm{GL}(n,2),\ c \in \mathbb{F}_2^n\}.$$

This includes matrices that are not scalar multiplication in a single field structure on $\mathbb{F}_2^n$.

Proposition 7.3. *For $n \geq 1$, put*

$$d_n = h_n \sum_{j=0}^{n} \frac{(-1)^j 2^{j(j-1)/2}}{h_j}, \qquad r_n = \begin{cases} 0, & n \text{ odd}, \\ h_n/|\mathrm{GL}(n/2,4)|, & n \text{ even}. \end{cases}$$

There are $2^n d_n$ affine complete mappings. Under F they form r_n singleton orbits and $(2^n d_n - r_n)/3$ orbits of length three. The fixed maps are exactly $x \mapsto Lx$ with $L^2 + L + I = 0$.

Proof. The number d_n counts invertible matrices with no nonzero fixed vector. Its displayed expression is the zero-dimensional fixed-space case of the classical Rudvalis–Shinoda count, recorded in [27, Theorem 2.1]. Here is a direct proof. For a j-dimensional subspace U put $\mu(U) = (-1)^j 2^{j(j-1)/2}$. If W has dimension r, the Gaussian-binomial identity gives

$$\sum_{U \leq W} \mu(U) = \sum_{j=0}^{r} \binom{r}{j}_2 (-1)^j 2^{j(j-1)/2} = \prod_{i=0}^{r-1}(1 - 2^i).$$

This is 1 when $W = 0$ and 0 otherwise. Apply it to $W = \ker(I+L)$ and sum over $L \in \mathrm{GL}(n,2)$. For a fixed j-subspace U, the matrices fixing U pointwise have block form $\left(\begin{smallmatrix} I & B \\ 0 & C \end{smallmatrix}\right)$, with $2^{j(n-j)}$ choices of B and h_{n-j} choices of C. Since there are $\binom{n}{j}_2$ choices of U and

$$\binom{n}{j}_2 2^{j(n-j)} h_{n-j} = h_n / h_j,$$

interchanging the two finite sums yields the formula for d_n. Each such L permits all 2^n translations c.

For $f(x) = Lx \oplus c$, the condition $F(f) = f$ is equivalent to $(\mathrm{id} \oplus f) \circ f = \mathrm{id}$. Comparing linear and constant parts gives $L^2 + L = I$ and $Lc = 0$, so $c = 0$. The polynomial $z^2 + z + 1$ is irreducible over $\mathbb{F}_2$. Thus such an L equips V with an $\mathbb{F}_4$-vector-space structure, which requires even n. Conversely, for even n multiplication by a root of this polynomial supplies such an L. All these matrices form one conjugacy class in $\mathrm{GL}(n,2)$: choose an $\mathbb{F}_4$-basis in each structure and compare the resulting $\mathbb{F}_2$-bases. The invertible maps commuting with L are exactly the $\mathbb{F}_4$-linear ones, so the centralizer has order $|\mathrm{GL}(n/2,4)|$. This proves the formula for r_n. Finally Lemma 4.2 restricts the other orbits to length three. □

The following small values are exact evaluations of the formulas, independently checked by enumerating all binary matrices and translations in these dimensions:

n	d_n	$\lvert\mathcal{C}_n^{\mathrm{aff}}\rvert$	F-fixed maps	3-element orbits
2	2	8	2	2
3	48	384	0	128
4	5824	93184	112	31024

Every F-fixed affine map is linear. In particular the 1680 nonlinear fixed maps in dimension four lie outside the affine class, showing the additional scope of Proposition 7.2.

Linear conjugacy of oriented spreads

We now distinguish the F-orbits of maps from equivalence under a linear change of coordinates. Write $f \sim g$ if $g = SfS^{-1}$ for some $S \in \mathrm{GL}(n,2)$. Such conjugacy preserves $F(f) = f$. For a spread $\mathcal{S}$, let $G_{\mathcal{S}}$ be its setwise stabilizer in $\mathrm{GL}(n,2)$, and choose a reference cyclic orientation on each line. An orientation is then a vector $(\epsilon_L)_{L\in\mathcal{S}}$ over $\mathbb{F}_2$, with 1 meaning reversal of the reference orientation. For $g \in G_{\mathcal{S}}$ define $\delta_g(L) \in \mathbb{F}_2$ by comparing the transported reference orientation on gL with the chosen reference on gL. Thus the transported bits satisfy

$$\epsilon'_{gL} = \epsilon_L \oplus \delta_g(L).$$

Proposition 7.4 (Orientation-sensitive orbit formula)**.** *The correspondence of Proposition 7.2 identifies linear conjugacy with equivalence of oriented spreads. Let $c(g)$ be the number of cycles in the permutation of spread lines induced by g, and put*

$$e(g) = \begin{cases} 1, & \bigoplus_{L\in C} \delta_g(L) = 0 \text{ for every line cycle } C, \\ 0, & \text{otherwise.} \end{cases}$$

The number of linear-conjugacy classes supported on the orbit of $\mathcal{S}$ is

$$a(\mathcal{S}) = \frac{1}{|G_{\mathcal{S}}|} \sum_{g\in G_{\mathcal{S}}} e(g) 2^{c(g)}.$$

Summing this over representatives of the linear equivalence classes of underlying spreads gives all F-fixed linear-conjugacy classes. The stabilizer of a fixed map is the stabilizer of its oriented spread.

Proof. Linear conjugacy sends each cycle $(x, f(x), x \oplus f(x))$ to its transported cycle, so it sends the spread and its orientations to exactly those of SfS^{-1}. To fix an orientation, the displayed bit relation must hold with $\epsilon' = \epsilon$. Around a line cycle the successive equations are consistent if and only if the sum of the $\delta_g(L)$ on that cycle is zero. When consistent, there are two choices for its first bit and all other bits are forced. Thus g fixes $e(g)2^{c(g)}$ orientations. Burnside's orbit-counting lemma gives the formula. Restricting to the stabilizer of one underlying spread suffices for its full linear orbit; distinct underlying spread orbits cannot give conjugate maps. The same transport correspondence gives the stabilizer assertion. □

Proposition 7.5 (Local criterion for an invariant orientation)**.** *Let $\mathcal{S}$ be any line spread and $G = G_{\mathcal{S}}$ its setwise stabilizer in $\mathrm{GL}(V)$. For $L \in \mathcal{S}$, write $G_L = \{g \in G : gL = L\}$. The following conditions are equivalent:*

1. *There is an orientation of every line in $\mathcal{S}$ that is fixed by all of G.*
2. *Every $g \in G_L$, for every $L \in \mathcal{S}$, induces an even permutation of the three nonzero vectors of L.*
3. *The orientation-translation cocycle associated with any choice of reference orientations is a coboundary.*

More precisely, put $E = \mathbb{F}_2^{\mathcal{S}}$ with its natural permutation G-module structure $(g \cdot v)_{gL} = v_L$. If $D_g \in E$ is defined by $(D_g)_{gL} = \delta_g(L)$, then $D_{gh} = D_g + g \cdot D_h$, and condition (3) means that there exists $v \in E$ with $D_g = v + g \cdot v$ for every $g \in G$.

Proof. An orientation of a line is a cyclic ordering of its three nonzero vectors, up to cyclic rotation. Its stabilizer within the permutation group on those vectors is the alternating group of degree three. Hence (1) implies (2).

Assume (2). Choose one representative L from each G-orbit on $\mathcal{S}$ and choose an arbitrary cyclic orientation on it. Transport the chosen orientation to gL using $g \in G$. If $gL = hL$, then $h^{-1}g \in G_L$ and (2) says that this element preserves the chosen orientation. Thus the transport is independent of the choice of g, and the resulting orientation on all lines is fixed by G. This proves (2)⇒(1).

For the last equivalence, the transported bit vector is $g \cdot v + D_g$. Composing two transports gives $D_{gh} = D_g + g \cdot D_h$. The bit vector v is fixed by every g precisely when $v = g \cdot v + D_g$, or equivalently $D_g = v + g \cdot v$. This is exactly the coboundary condition. □

The proposition distinguishes the existence of one orientation fixed by the entire spread automorphism group from the individual fixed-orientation tests $e(g)$ in Proposition 7.4; the latter still supply the general orbit count even when no globally invariant orientation exists.

Theorem 7.6 (Four-dimensional classification)**.** *There are exactly three conjugacy classes under* $\mathrm{GL}(4,2)$ *of F-fixed complete mappings of $\mathbb{F}_2^4$. Identify V with $\mathbb{F}_4^2$, choose $\omega^2 + \omega + 1 = 0$, and start with $f_0(x) = \omega x$ on its five spread lines. Representatives are obtained by reversing any r lines, with $r = 0, 1, 2$. Their orbit sizes, stabilizer orders and stabilizer groups are*

r	*orbit size*	*stabilizer order*	*stabilizer group*	*linear*
0	*112*	*180*	$C_3 \times A_5$	*yes*
1	*560*	*36*	$C_3 \times A_4$	*no*
2	*1120*	*18*	$C_3 \times S_3$	*no*

Here C_3 is cyclic of order three, and A_j, S_j denote alternating and symmetric groups. In particular the 1680 *nonlinear maps form two classes.*

Proof. First every four-dimensional spread is equivalent to the displayed one. Choose two spread lines as complementary spaces $E \oplus E$, where $E = \mathbb{F}_2^2$. The other three lines are graphs of invertible maps $P_i : E \to E$, with all $P_i + P_j$ invertible. A change of coordinates makes one of these maps I. Among the six matrices of $\mathrm{GL}(2,2)$, the only ones for which $I + P$ is invertible are

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad P^2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

They satisfy $P^2 + P + I = 0$. The three graphs are consequently those of I, P, P^2, giving the five one-dimensional $\mathbb{F}_4$-subspaces of $\mathbb{F}_4^2$.

For this standard spread, the only linear maps inducing cyclic orientations on its lines are f_0 and f_0^2. Indeed on each line such a map is multiplication by ω or ω^2; additivity at $e_1 + e_2$ forces the choices on the two coordinate lines to agree, which determines the map on their direct sum. Therefore any linear stabilizer of the underlying spread conjugates f_0 to either f_0 or f_0^2. The commuting maps are $\mathrm{GL}(2,4)$; composing with the coordinatewise Frobenius map $\sigma(x_1, x_2) = (x_1^2, x_2^2)$ accounts for the other coset. Thus the full spread stabilizer is $\Gamma\mathrm{L}(2,4)$ of order $2|\mathrm{GL}(2,4)| = 360$. Since $|\mathrm{GL}(4,2)| = 20160$, there are $20160/360 = 56$ spreads, recovering the classical count [3].

On the five lines, $\mathrm{GL}(2,4)$ has scalar kernel C_3 and image $\mathrm{PGL}(2,4)$ of order 60. This index-two subgroup of S_5 is A_5. Its elements preserve every reference orientation. Frobenius reverses them all and induces the transposition exchanging ω and ω^2 on $\mathbb{P}^1(\mathbb{F}_4)$, fixing $0, 1, \infty$. Consequently even line permutations permute the reversed-line subset, whereas odd line permutations also complement it. A_5 is transitive on subsets of each fixed size: an arbitrary permutation carrying one subset to

another can have its parity adjusted by a transposition inside the subset or its complement. Hence the orientation orbits are exactly the weight pairs $\{0,5\}$, $\{1,4\}$, $\{2,3\}$, of sizes $2, 10, 20$. Multiplying by the 56 underlying spreads gives the three displayed orbit sizes and orbit–stabilizer gives their stabilizer orders.

For the representatives of weights $r = 0, 1, 2$, an odd line permutation cannot stabilize the orientation because it changes the weight to $5-r$. Within $\mathrm{GL}(2,4)$ the projective stabilizers are A_5, A_4, and the two-subset stabilizer S_3, respectively. The determinant splits $\mathrm{GL}(2,4) = C_3 \times \mathrm{SL}(2,4)$; the projective action identifies $\mathrm{SL}(2,4)$ with A_5, giving the displayed direct products. The earlier two-linear-map observation shows that only the first class is linear. □

The proof gives representatives and stabilizers, rather than only the total number of maps. Independently, `verify_affine_groups_and_conjugacy.py` constructs all 1792 maps and conjugates the three representatives by all 20160 matrices; their disjoint orbits exhaust the list. It also compares direct fixed-orientation counts with Proposition 7.4 for all 360 standard-spread automorphisms. For dimension six, underlying spreads have already been classified by Mateva and Topalova [28] into 131044 types. We next treat the standard Desarguesian type, one of these types. The other underlying types are outside the six-dimensional computation reported here.

Standard spreads in arbitrary even dimension

Let $V = \mathbb{F}_4^m$, let $\omega^2 + \omega + 1 = 0$, and let $\mathcal{S}_m$ consist of its one-dimensional $\mathbb{F}_4$-subspaces, viewed as two-dimensional $\mathbb{F}_2$-subspaces. Put

$$N_m = \frac{4^m - 1}{3}, \qquad P_m = \mathrm{PGL}(m,4), \qquad p_m = |P_m| = \frac{1}{3}\prod_{i=0}^{m-1}(4^m - 4^i).$$

Orient each line by $x \mapsto \omega x$ before choosing the reversed-line subset. Write a_m for the number of full linear-conjugacy classes supported on spreads equivalent to $\mathcal{S}_m$.

Theorem 7.7 (Standard-spread action and asymptotics)**.** *For $m \geq 2$, the setwise stabilizer of $\mathcal{S}_m$ in $\mathrm{GL}(2m,2)$ is $\Gamma\mathrm{L}(m,4)$. Its linear coset permutes orientation bits; its semilinear coset also complements every bit and fixes no orientation. Every oriented stabilizer contains the scalar group $C_3 = \mathbb{F}_4^*$ and is contained in $\mathrm{GL}(m,4)$. If $c(g)$ counts the cycles of g on the N_m projective points, then*

$$a_m = \frac{1}{2p_m}\sum_{g \in P_m} 2^{c(g)} = \frac{2^{N_m}}{2p_m}(1 + \varepsilon_m), \qquad 0 \leq \varepsilon_m \leq (p_m - 1)2^{-2^{2m-3}}. \tag{39}$$

For a uniformly chosen orientation,

$$\Pr\big(\mathrm{Stab}_{\mathrm{GL}(2m,2)}(f) \neq C_3\big) \leq (p_m - 1)2^{-2^{2m-3}}. \tag{40}$$

Consequently almost every orientation has stabilizer exactly C_3 as $m \to \infty$, and

$$a_m \sim \frac{3\,2^{(4^m-1)/3}}{2|\mathrm{GL}(m,4)|}.$$

Such orientations exist for every $m \geq 4$ and are nonlinear. Furthermore, exactly $a_m - 1$ of the a_m supported conjugacy classes are nonlinear for every $m \geq 2$.

Proof. A linear map giving cyclic orientations on the lines of $\mathcal{S}_m$ must be either $f_0(x) = \omega x$ or $f_0^2(x) = \omega^2 x$. On each line it has one of these two restrictions. Applying additivity to independent coordinate vectors and their sum forces their choices to agree, and the two $\mathbb{F}_2$-basis vectors on each coordinate line then determine the map on V. A spread stabilizer conjugates f_0 to one of these two maps. The centralizer of f_0 is $\mathrm{GL}(m, 4)$; coordinatewise Frobenius $\sigma(x) = x^2$ conjugates f_0 to f_0^2. This proves the spread-stabilizer assertion.

Linear elements preserve the reference orientations; elements of $\mathrm{GL}(m, 4)\sigma$ reverse all of them. Such an element sends an orientation of weight r to one of weight $N_m - r$. As N_m is odd, it fixes none. Scalars act trivially on the orientation set, giving the first equality in (39) by Proposition 7.4. The orientation stabilizer is therefore the inverse image in $\mathrm{GL}(m, 4)$ of its subset stabilizer in P_m.

For a nonscalar matrix A, its fixed projective points are the lines in its eigenspaces for the three nonzero elements of $\mathbb{F}_4$. If their dimensions are d_λ, the number is $\sum_\lambda (4^{d_\lambda} - 1)/3$. Their dimensions sum to at most m and each is at most $m-1$. Convexity gives the upper bound $(4^{m-1}-1)/3+1$. Thus every nonidentity projective element moves at least $4^{m-1} - 1$ points. Each nontrivial permutation cycle has at least two points, so

$$N_m - c(g) \geq \left\lceil \frac{4^{m-1} - 1}{2} \right\rceil = 2^{2m-3}.$$

It fixes a proportion $2^{c(g)-N_m}$ of all orientations. Summing these proportions bounds both the nonidentity contribution in (39) and, by the union bound, the probability in (40). This is the classical motion argument [30] in the present projective action.

Finally $p_m < 2^{2m^2}/3$ and $2^{2m-3} \geq 2m^2$ for $m \geq 4$; the latter holds at $m = 4$ and increases thereafter. The exceptional bound is then less than one and tends to zero. The only two linear orientations have stabilizer $\mathrm{GL}(m, 4)$, which is larger than C_3, so orientations with exactly C_3 are nonlinear. The two uniform orientations are conjugate by the semilinear Frobenius map, and no other orientation is linear; hence they constitute the single linear conjugacy class and the other $a_m - 1$ classes are nonlinear. □

For example $p_4 = 987033600$, and the exceptional proportion is at most $987033599/2^{32} < 0.230$. The estimate therefore already guarantees that more than 77% of the orientations on the standard eight-dimensional spread have exactly the scalar stabilizer. This conclusion concerns the standard spread family, without asserting a distribution over all spreads.

Corollary 7.8 (Six-dimensional standard-spread classes)**.** *The 2^{21} orientations of $\mathcal{S}_3$ give exactly* 80 *full* $\mathrm{GL}(6, 2)$ *conjugacy classes supported on its underlying spread orbit. Classifying by the smaller of the two complementary orientation weights, the numbers are*

r	0	1	2	3	4	5	6	7	8	9	10
classes	1	1	1	2	3	5	8	10	13	17	19.

Only the weight-zero class is linear.

Proof. Here $|P_3| = 60480$. Its projective cycle types and multiplicities are listed below; a superscript is the number of cycles of that length.

matrix description	projective cycle type	multiplicity
scalar	1^{21}	1
$J_2(a) \oplus a$	$1^5\,2^8$	315
$J_3(a)$	$1\,2^2\,4^4$	3780
$aI_2 \oplus b$, $a \neq b$	$1^6\,3^5$	672
$J_2(a) \oplus b$, $a \neq b$	$1^2\,2^2\,3\,6^2$	10080
three distinct eigenvalues	$1^3\,3^6$	2240
quadratic block, normalized order 5	$1\,5^4$	8064
quadratic block, normalized order 15	$1\,5\,15$	16128
cubic block, projective order 3	3^7	1920
cubic block, projective order 7	7^3	5760
cubic block, projective order 21	21	11520

For completeness these multiplicities follow directly from rational canonical form. In $\mathrm{GL}(3,4)$, the first six types have respectively $3, 3, 3, 6, 6, 1$ conjugacy classes. Their corresponding centralizer orders are

$$181440, \quad 576, \quad 48, \quad 540, \quad 36, \quad 27.$$

Divide their total element counts by three to pass to projective matrices. For a quadratic block plus eigenvalue a, divide the quadratic roots by a. The 12 elements of $\mathbb{F}_{16} \setminus \mathbb{F}_4$ have orders 5 or 15, with 4 or 8 such roots. They give 6 or 12 matrix classes with centralizer order 45. For a cubic block, roots outside $\mathbb{F}_4$ have projective orders $3, 7, 21$, represented by $6, 18, 36$ field elements. They give $2, 6, 12$ matrix classes with centralizer order 63. For the Jordan types, $J_2(a) \oplus a$ has projective order two and five fixed points. The powers of $J_3(a)$ have respectively one, five, and twenty-one fixed points at exponents $1, 2, 4$, giving $1\,2^2\,4^4$. For $J_2(a) \oplus b$, the corresponding counts at exponents $1, 2, 3, 6$ are $2, 6, 5, 21$, giving $1^2\,2^2\,3\,6^2$. The remaining cycle types follow from the orders on the invariant line, quadratic component, or cubic field. All multiplicities sum to 60480, so the types are exhaustive.

Summing multiplicity times $2^{\text{number of cycles}}$ gives 9676800; (39) gives $9676800/120960 = 80$. The number of r-subset classes under P_3 is the coefficient of z^r in

$$\frac{1}{60480} \sum_{g \in P_3} \prod_{C \text{ a point cycle of } g} (1 + z^{|C|}).$$

Using the table gives the displayed row. Semilinear conjugation pairs weights r and $21 - r$, so each entry also counts the oriented classes of the corresponding weight pair. Only the two uniform orientations are linear, and they belong to the same class. □

An independent complete traversal of the 2^{21} orientation vectors supplies all 80 representatives, their full linear stabilizer orders, and their orbit sizes. Among these, four classes have stabilizer C_3, accounting for 483840 orientations on the fixed standard spread. These representative and stabilizer data are finite computations, distinct from the general proof of Theorem 7.7. The supplied verifier also independently enumerates the 60480 projective matrices and recovers the preceding cycle table and weight polynomial.

Coordinate remarks and scope

For the six-bit example, the fixed point 38 can serve as the origin after relabeling, while 0 remains the additive origin. In the value set $\{0, \ldots, 63\}$ the 9-multiples are exactly the 8 self-doubled words $(x_i = x_{i+3})$, a 3-dimensional subspace. To parametrize the long orbit, define

$$\theta(j) = T^j(63), \qquad j \in \mathbb{Z}/63.$$

This is a bijection onto $V \setminus \{38\}$. Apply the CRT isomorphism to the *iteration exponent* j, sending it to $(j \bmod 7, j \bmod 9)$. Thus $\theta(0) = 63$ corresponds to $(0,0)$, whereas $0 = \theta(62)$ corresponds to $(6,8)$, since $T(0) = 63$. Reduction of the integer word labels modulo 63 does not give this parametrization. The cyclotomic factor Φ_{2^k+1} reflects $(2^k-1)(2^k+1) = 2^{2k}-1$; the fixed point of K accounts for the remaining point.

The general fixed-map description reduces counting to the line-spread enumeration problem; no general formula for s_n is supplied here. The affine orbit count is complete for every dimension, while the scalar Singer triples provide a distinguished family with controlled permutation cycle types and two-generator affine groups. Theorems 5.1, 7.6 and 7.7 give the general affine group structure, the full four-dimensional classification, and the standard-spread asymptotics. Corollary 7.8 counts all six-dimensional standard-spread oriented classes; it does not classify orientations on the other 131043 underlying spread types (out of 131044 in total). The next section treats nonlinear companion groups.

8 Nonlinear companion groups

An F-fixed map generates only a cyclic group of order three with its companion. Large companion groups therefore require a three-element F-orbit. If two companions both have cycle type $1+(2^n-1)$, their cycles have odd length, so their group is contained in A_{2^n}. The following example attains this upper bound in dimension four.

Proposition 8.1 (Two long cycles with distinct fixed points)**.** *Let Ω be a finite set of size $N \geq 4$, and suppose $f, g \in \mathrm{Sym}(\Omega)$ both have cycle type $1+(N-1)$. If their unique fixed points are distinct, then $H = \langle f, g \rangle$ acts doubly transitively on Ω. If in addition $N = 2^n$ with $n \geq 3$ and H contains a 3-cycle, then $H = A_N$.*

Proof. Let a be the unique fixed point of f. Since g fixes a different point, $g(a) \neq a$. The $(N-1)$-cycle f acts transitively on $\Omega \setminus \{a\}$, so H is transitive on Ω: its orbit containing a also contains $g(a)$ and every other point. Moreover $f \in H_a$, and f itself is transitive on $\Omega \setminus \{a\}$. Thus the point stabilizer is transitive on the remaining points and H is doubly transitive.

Now let $N = 2^n \geq 8$. An $(N-1)$-cycle is even, so $H \leq A_N$. The action is primitive since it is doubly transitive. The classical Jordan theorem for a primitive degree-N group containing a prime cycle of length $p \leq N-3$, applied with $p = 3$, implies $A_N \leq H$ (see, e.g., [19]). Therefore $H = A_N$. □

This sufficient condition separates the group-theoretic deduction from the difficult existence problem for complete mappings with prescribed companion cycle types: the distinct-fixed-point hypothesis gives double transitivity automatically, while the existence of a suitable short-cycle element still requires additional arguments.

Proposition 8.2 (A nonlinear alternating-group example)**.** *On $V = \mathbb{F}_2^4$, with integer labels and XOR addition, let f be the permutation in the first row below and let $g = F(f)$.*

x	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$f(x)$	1	5	12	3	11	7	15	10	2	14	9	13	0	8	6	4
$g(x)$	3	0	5	10	1	13	11	9	14	6	8	15	12	7	2	4

Then f is complete and nonlinear. Each of $f, F(f), F^2(f)$ has cycle type $1+15$, with fixed points $3, 12, 1$ respectively, and

$$\langle f, F(f) \rangle = A_{16}.$$

Proof. The first row and $x \oplus f(x)$ each permute the labels $0, \dots, 15$; inverting the latter gives the second row. The cycles other than the indicated fixed points are

$$f : (0\ 1\ 5\ 7\ 10\ 9\ 14\ 6\ 15\ 4\ 11\ 13\ 8\ 2\ 12),$$
$$g : (0\ 3\ 10\ 8\ 14\ 2\ 5\ 13\ 7\ 9\ 6\ 11\ 15\ 4\ 1),$$
$$F^2(f) : (0\ 12\ 14\ 7\ 2\ 10\ 13\ 6\ 8\ 5\ 4\ 11\ 15\ 9\ 3).$$

The third cycle follows from $F^2(f) = \mathrm{id} \oplus f^{-1}$. An affine map would satisfy $f(1 \oplus 2) = f(1) \oplus f(2) \oplus f(0)$. Here these values are 3 and 8, so f is nonlinear. Since F preserves the affine class, all three companions are nonlinear.

Use right-to-left composition, write $[u, v] = uvu^{-1}v^{-1}$, and put $w = f^2 g$. Directly from the table,

$$w = (0\ 3\ 14)(1\ 5\ 2\ 10\ 12)(4\ 7\ 6\ 8\ 15\ 13\ 9), \qquad w(11) = 11.$$

Thus $s = w^{35} = (3\ 0\ 14)$. All generators are even, so $H = \langle f, g \rangle \leq A_{16}$. The cycles $f^j s f^{-j} = (3\ f^j(0)\ f^j(14))$ belong to H, and their supports contain every point because f cycles all points other than 3.

To see that these cycles generate A_{16}, make a graph on the other 15 points, joining a, b whenever $(3\ a\ b)$ is available. There are no isolated vertices. If two graph components have edges ab and cd, the identity $[(3\ a\ b), (3\ c\ d)] = (3\ a\ c)$ joins them. After the graph is connected, conjugating $(3\ b\ c)$ by $(3\ a\ b)$ gives $(a\ 3\ c)$, so paths supply $(3\ a\ b)$ for every pair. Finally $(3\ a\ b)(3\ b\ c) = (a\ b\ c)$. These are all three-cycles, which generate A_{16}. Hence $H \geq A_{16}$, proving equality without a group-order search. □

Alternatively, the distinct fixed points 3 and 12 give double transitivity by Proposition 8.1; the explicit word w^{35} is a 3-cycle. The Jordan-theorem conclusion in that proposition therefore yields A_{16} directly. The elementary proof above is retained independently of Jordan's theorem.

Complete mapping polynomials over $\mathbb{F}_{16}$ and their reduced-degree distribution were computationally studied by Yuan–Tong–Zhang [34]; Bors–Wang studied cycle types and coset-wise affine constructions of complete mappings [25, 33]. These works provide important precedents for the finite-field maps and cycle-type methods. The result here concerns the additional group generated by one specified map and its prescribed companion, which is not the object counted by those cited results. A construction in infinitely many dimensions with simultaneous cycle-type and companion-group control is not supplied here. The groups generated by the full collections of complete mappings in [26] concern a different set of generators.

Proposition 8.3 (Translations of fixed maps)**.** *Let f_0 be an F-fixed complete mapping on $\mathbb{F}_2^n$ and let $b \neq 0$. Write $\tau_b(x) = x \oplus b$ and $f = \tau_b f_0$. Then f is complete and*

$$F(f) = f_0 \tau_b, \qquad F^2(f) = \tau_b f_0 \tau_b, \qquad \langle f, F(f) \rangle = \langle f_0, \tau_b \rangle.$$

The orbit of 0 *under this group is exactly the four-point line*

$$U_b = \{0, b, f_0(b), f_0^2(b)\}.$$

The induced action on U_b is A_4. Thus this construction cannot give a transitive companion group when $n > 2$.

Proof. The fixed-map identities give $f_0^3 = \mathrm{id}$ and $\mathrm{id} \oplus f_0 = f_0^2$. Therefore $\mathrm{id} \oplus f = \tau_b f_0^2$ is a permutation, and its inverse is $g = f_0 \tau_b$. Also

$$(\mathrm{id} \oplus g)(x) = x \oplus f_0(x \oplus b) = b \oplus f_0^2(x \oplus b),$$

so taking its inverse gives the formula for $F^2(f)$. Since $gf = f_0^2$, the group generated by f, g contains f_0 and then $\tau_b = f f_0^{-1}$; the reverse inclusion is immediate.

Proposition 7.2 makes U_b a two-dimensional subspace, invariant under both f_0 and τ_b. Starting at 0, translation reaches b and f_0 reaches the other two nonzero points, so the orbit is all of U_b. On these points $f_0 = (b\ f_0(b)\ f_0^2(b))$ and $\tau_b = (0\ b)(f_0(b)\ f_0^2(b))$. These generate the three double transpositions and a three-cycle, hence A_4. □

References

[1] M. Hall, L. J. Paige, *Complete mappings of finite groups*, Pacific J. Math. 5 (1955) 541–549, doi:10.2140/pjm.1955.5.541.

[2] B. D. McKay, I. M. Wanless, *Enumeration of Latin squares with conjugate symmetry*, J. Combin. Des. 30 (2022) 105–130, doi:10.1002/jcd.21814.

[3] N. Magaud, *Proof Pearl: Formalizing Spreads and Packings of the Smallest Projective Space PG(3,2) Using the Coq Proof Assistant*, in ITP 2022, LIPIcs 237 (2022) 25:1–25:17, doi:10.4230/LIPIcs.ITP.2022.25.

[4] R. Lidl, H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge, 1997.

[5] P. van Wamelen, *Jacobi sums over finite fields*, Acta Arith. 102(1) (2002) 1–20, doi:10.4064/aa102-1-1.

[6] T. K. Lim, C. E. Praeger, *On generalized Paley graphs and their automorphism groups*, Michigan Math. J. 58(1) (2009) 293–308, doi:10.1307/mmj/1242071694.

[7] M. L. Dawsey, D. McCarthy, *Generalized Paley graphs and their complete subgraphs of orders three and four*, Res. Math. Sci. 8 (2021), Article 18, arXiv:2006.14716.

[8] R. A. Podestá, D. E. Videla, *Spectral properties of generalized Paley graphs of $(q^\ell + 1)$-th powers and applications*, arXiv:1812.03332v7 (2024).

[9] N. M. Katz, *An estimate for character sums*, J. Amer. Math. Soc. 2 (1989) 197–200, author's copy.

[10] R. R. Farashahi, P.-A. Fouque, I. E. Shparlinski, M. Tibouchi, J. F. Voloch, *Indifferentiable deterministic hashing to elliptic and hyperelliptic curves*, Math. Comp. 82 (2013) 491–512, doi:10.1090/S0025-5718-2012-02606-8.

[11] S. D. Cohen, G. Kapetanakis, *Finite field extensions with the line or translate property for r-primitive elements*, J. Aust. Math. Soc. 111 (2021) 313–319, doi:10.1017/S1446788720000099.

[12] S. D. Cohen, G. Kapetanakis, L. Reis, *The existence of $\mathbb{F}_q$-primitive points on curves using freeness*, C. R. Math. 360 (2022) 641–652, doi:10.5802/crmath.328.

[13] J. J. R. Aguirre, V. G. L. Neumann, *Pairs of r-primitive and k-normal elements in finite fields*, arXiv:2210.11504v1 (2022), arXiv:2210.11504.

[14] D. Hachenberger, *Ovoids and primitive normal bases for quartic extensions of Galois fields*, J. Algebr. Comb. 53 (2021) 85–114, doi:10.1007/s10801-019-00920-8.

[15] J. Sharma, R. Sarma, S. Laishram, *Arithmetic progressions of r-primitive elements in a field*, Bull. Braz. Math. Soc. (N.S.) 55 (2024), Article 40, doi:10.1007/s00574-024-00412-9.

[16] K. Chatterjee, H. Sharma, A. Shukla, S. K. Tiwari, *Arithmetic progression in a finite field with prescribed norms*, Forum Math. 37(1) (2025) 345–357, doi:10.1515/forum-2024-0026.

[17] A. Choudhary, J. Sharma, *Arithmetic progressions of some special elements in finite fields*, Commun. Algebra (published online 2025), doi:10.1080/00927872.2025.2561951.

[18] H. Kudo, Y. Nogata, *Cyclotomic numbers of order $q-1$ over $\mathbb{F}_{q^r}$*, arXiv:2604.25461 (2026), `https://arxiv.org/abs/2604.25461`.

[19] J. D. Dixon, B. Mortimer, *Permutation Groups*, Springer, 1996.

[20] S. D. Cohen, *The orders of related elements of a finite field*, Ramanujan J. 7 (2003) 169–183, doi:10.1023/A:1026295128600.

[21] O. Moreno, *On the existence of a primitive quadratic of trace* 1 *over* $GF(p^m)$, J. Combin. Theory Ser. A 51 (1989) 104–110.

[22] S. D. Cohen, *Primitive elements and polynomials with arbitrary trace*, Discrete Math. 83 (1990) 1–7.

[23] S. D. Cohen, M. Prešern, *Primitive finite field elements with prescribed trace*, Southeast Asian Bull. Math. 29(2) (2005) 283–300, publication record.

[24] C. Gravel, D. Panario, D. Thomson, *Unicyclic strong permutations*, arXiv:1809.03551 (2018); Cryptogr. Commun. 11 (2019) 1211–1231.

[25] A. Bors, Q. Wang, *Cycle types of complete mappings of finite fields*, J. Algebra 591 (2022) 577–610, doi:10.1016/j.jalgebra.2021.09.017.

[26] A. Bors, Q. Wang, *Compositions and parities of complete mappings and of orthomorphisms*, J. Combin. Theory Ser. A 196 (2023) 105723, doi:10.1016/j.jcta.2022.105723.

[27] J. Fulman, D. Stanton, *On the distribution of the number of fixed vectors for the finite classical groups*, Ann. Comb. 20 (2016) 755–773, doi:10.1007/s00026-016-0336-7.

[28] Z. T. Mateva, S. T. Topalova, *Line spreads of PG(5,2)*, J. Combin. Des. 17(1) (2009) 90–102, doi:10.1002/jcd.20198.

[29] A. Müyesser, *Cycle type in Hall–Paige: a proof of the Friedlander–Gordon–Tannenbaum conjecture*, Forum Math. Sigma 14 (2026) e50, doi:10.1017/fms.2026.10197.

[30] A. Russell, R. Sundaram, *A note on the asymptotics and computational complexity of graph distinguishability*, Electron. J. Combin. 5(1) (1998), Research Paper 23, doi:10.37236/1361.

[31] P. J. Cameron, P. M. Neumann, J. Saxl, *On groups with no regular orbits on the set of subsets*, Arch. Math. 43 (1984) 295–296, doi:10.1007/BF01196649.

[32] Á. Seress, *Primitive groups with no regular orbits on the set of subsets*, Bull. London Math. Soc. 29 (1997) 697–704, doi:10.1112/S0024609397003536.

[33] A. Bors, Q. Wang, *Coset-wise affine functions and cycle types of complete mappings*, Finite Fields Appl. 83 (2022) 102088, doi:10.1016/j.ffa.2022.102088.

[34] Y. Yuan, Y. Tong, H. Zhang, *Complete mapping polynomials over finite field* $\mathbb{F}_{16}$, in *Arithmetic of Finite Fields*, WAIFI 2007, Lecture Notes in Computer Science 4547, Springer (2007), 147–158, doi:10.1007/978-3-540-73074-3_12.

A Computational appendix

All numerical claims (fixed points, cycle types, group order 4032, and the characteristic polynomials of L_T, L_D, and L_K) were verified computationally. The script `verify_paperA_structure.py` reproduces the coordinate, cycle, group-order, basis-ordering, and τ-collapse data, as well as the $\mathbb{F}_{64}$ instance of Proposition 7.1 and the CRT exponent convention. It computes the characteristic polynomials as $\det(tI + L)$ by the permutation expansion in characteristic two. The separate `verify_k1_29.py` checks the finite PTE certificate. The script `verify_complete_mapping_orbits.py` enumerates all binary matrices and translations for $n \leq 4$, checks Lemmas 4.1–4.2 and the coordinate-permutation identities of Proposition 4.3, and compares the resulting counts with Proposition 7.3. For $n = 4$ it enumerates line spreads by exact cover, orients every line, checks $F(f) = f$, and tests additivity, reproducing the counts following Proposition 7.2. The additional `verify_affine_groups_and_conjugacy.py` partitions all admissible matrices for $n \leq 4$ into linear-conjugacy classes and directly closes the two-generator permutation group for every translation at each matrix representative. The $4, 16, 80$ closure cases for $n = 2, 3, 4$, respectively, cover all $8, 384, 93184$ affine pairs by linear conjugacy. These closures are compared with Theorem 5.1; transitivity, block systems, and double transitivity are also tested directly. The same script checks Theorem 7.6 using all 1792 maps and all 20160 linear changes of coordinates, and verifies the signed Burnside count. It shares basic permutation and spread routines with `verify_complete_mapping_orbits.py`; the direct closure computation does not use the proposed group formula. Its data file `affine_groups_and_conjugacy_results.json` includes every matrix representative and translation case, map representatives, stabilizer matrices, and the fixed-orientation counts. These deterministic scripts require only the Python standard library and were run with Python 3.12.14. The ancillary data include the 64 rows of T, D, K, all 720 basis orderings, the 29 PTE certificates with their factorizations, the 56 four-dimensional spreads, and a nonlinear fixed-map witness. The script `export_paperA_data.py` regenerates the coordinate and certificate tables; it reuses the two original verifiers rather than constituting an independent implementation. The scripts, data, proof sources, reproduction instructions, and the current SHA-256 manifest accompany this manuscript as ancillary files. The single command `pythonreproduce_paperA.py` checks the manifest, runs all finite verifiers, regenerates the data, and checks their hashes again. The project repository is

`https://github.com/63331849121/yijing-mathematical-structures`.

Relative to the standard basis $e_1, \ldots, e_6$, the linear part of the main map is

$$L_T = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{pmatrix}.$$

Formalization. `CompleteMappingTriality.lean` checks the function-level rotation, $F^2(f) = \mathrm{id} \oplus f^{-1}$, $F^3(f) = f$, and the unique-fixed-point assertion of Lemmas 4.1–4.2 for additive groups of exponent two.

The affine group-generation argument (for a field $\mathbb{K}$ with a primitive element α and $c \neq 0$, the maps T and D generate $\mathrm{AGL}(1, \mathbb{K})$, the core of Theorem 5.26 and §6) and the algebraic lemmas used in it (the commutator is a translation; conjugation of translations by T) have been formalized and machine-checked in `AGL.lean`.

The conditional file `CohenEvenConditional.lean` formalizes the field-level Frobenius identities and the resulting order-theoretic τ-collapse under an explicit order interface; it does not formalize the paper proof of the existence theorem. The concrete cycle-type data of §3–4 were verified by independent computation. The existence part of Theorem 6.3 is proved in Appendix B; its analytic part is not formalized in Lean, and its finite range is discharged by the deterministic certificate there. Thus Cohen's theorem is not used as a proof dependency; the classical references are cited only for historical attribution. The general matrix-group structure, its action criteria, the signed Burnside formula, four-dimensional conjugacy classification, coordinate-permutation description, line-spread classification, and affine orbit-count proof are paper proofs, not Lean proofs. The inverse-realization and arithmetic-sieve criteria, the primitive-group nonrealizability theorem, the positive Jacobi-sum/sieve bound, the fixed-index realization theorem and index-three threshold, the quantitative density and growing-index results, norm descent and the quarter-power obstruction, primitive inverse classification and the generating-triangle correspondence, the local invariant-orientation criterion, and the long-cycle double-transitivity criterion are also paper proofs, not part of the Lean formalization.

All three Lean files were checked using Lean 4.31.0 and mathlib commit `fabf563a7c95a166b8d7b6efca11c8b4dc9d911f`. The ancillary `lean/` directory contains these files, `lean-toolchain`, `lakefile.lean`, and `lake-manifest.json`, pinning the proof environment. These file checks used an existing dependency cache; a fresh dependency download and cloud CI run are not claimed. Code is supplied under the MIT license, and the manuscript and data under CC BY 4.0.

The verifier `verify_standard_spread_orbits.py` independently enumerates the 60480 projective matrices for the standard six-dimensional spread and traverses all 2^{21} orientations using elementary matrix generators and the signed Frobenius action. It checks the projective cycle table, the weight polynomial, and the 80 disjoint orientation orbits. The data `standard_spread_results.json` contain every representative, its map, full linear stabilizer order, and both its orientation-orbit size and its full $\mathrm{GL}(6, 2)$ conjugacy-class size. Only this underlying spread type is covered. The verifier `verify_nonlinear_companions.py` checks the explicit sixteen-point tables, all three cycle types, the three-cycle word certificate, and the translated-fixed identities. It does not enumerate A_{16}: the group equality has the elementary proof in Proposition 8.2. The arbitrary-dimensional asymptotic and translated-fixed results are paper proofs, outside the supplied Lean formalization.

The verifier `verify_scalar_realization.py` compares two independent finite counting methods: binary-polynomial gcds and explicit multiplicative subgroups in finite-field coordinates. It covers every divisor of $2^n - 1$ for $n = 2, 4, 6, 8, 10, 12, 14$, checks the Jacobi bounds and slope-generation sieve, and verifies the degree-eight witness in Corollary 5.8. The data `scalar_realization_results.json` include the index-three counts $|\mathcal{R}_d| = 0, 0, 24, 90, 396, 1722$ for $n = 4, 6, 8, 10, 12, 14$, respectively. These computations verify finite arithmetic certificates; the infinite-family and threshold assertions use the paper proofs in Section 5.

The verifier `verify_scalar_depth.py` audits the quantitative main term and error by exact rational arithmetic in even dimensions 2 through 14. It checks norm descent by polynomial gcds for $2 \leq r \leq 8$, all proper divisors $s \mid r$, and all $u \mid 2^s - 1$. The quarter-power counts are checked for $s = 1, 2, 3, 4, 5$ by polynomial gcds and the divisor sieve, with independent field enumeration and subfield-location checks for $s = 1, 2, 3, 4$. For primitive subgroup orders in even dimensions 2 through 14, it checks Frobenius and six-ratio orbits, the degree of the polynomial encoding the realizing slopes, and the conjugacy-class counts. It also directly enumerates generating triangles and

their affine stabilizers on 4 and 16 vertices. The corresponding data are `scalar_depth_results.json`. These are finite arithmetic audits; the uniform exponent obstruction and classification use the paper proofs.

Finite certificates for the inner critical theorem

Theorem 5.20 uses the analytic argument for $s \geq 33$ and the following complete finite certificate for $2 \leq s \leq 32$. A hexadecimal integer encodes a binary polynomial with its least significant bit the constant coefficient. Each row means

$$E = \mathbb{F}_2[X]/(M_s), \qquad L = E[Z]/(Z^2 + (1 + b + w)Z + b), \qquad a = Z.$$

Thus only M_s, b, w need be specified; all field operations are exact.

s	M_s (hex)	b (hex)	w (hex)
2	13	8	1
3	43	6	3b
4	11b	2f	1
5	409	9b	2c9
6	1009	c78	8e5
7	4021	3e8a	1
8	1002b	2878	689
9	40009	303be	12d39
10	100009	a3085	1
11	400003	662cc	3205f9
12	100001b	d0842d	1
13	400001b	32f9317	1
14	10000003	15a3baa	6c6a1f9
15	40000003	317d6a85	36559252
16	10000008d	4893e13d	2e37b719
17	40000001b	dcbd299	91607c0b
18	1000000035	78c0283c4	1
19	4000000063	1153a3d1ef	16004f10d3
20	10000000039	1f9215dca7	1
21	40000000027	2657c0cd035	374caba3695
22	100000000021	fe622270c1c	1
23	400000000003	17c45ffbd76	29bd79b1ae4
24	100000000002d	fed891dbe6c9	ec7a6a1920af
25	400000000001d	2907b6c341c09	1
26	10000000000009	c6d27ccd2b0df	432bf4829f702
27	4000000000007d	28047cef28c34e	1
28	100000000000095	a586528c5b1c9	1
29	400000000000063	243f16ec4e21119	1
30	1000000000000003	8c388b10c413f59	1
31	4000000000000069	12dfcd22c3b1728b	1
32	10000000000000001b	ae3111b3bb81428f	c171e75cc47c2b2e

The ancillary file `inner_critical_certificates.json` also records the full factorizations of A, B, every nonidentity residue $a^{d/p}$ for a prime $p \mid d$, and recursive Lucas primality certificates for those primes. Verification consists of the following finite identities, carried out by `verify_inner_critical.py`. The additional `verify_inner_symmetry.py` checks the four-torsion identities and

rational Lang-fiber multiplier at all 31 certificate points, and enumerates every primitive-norm fiber for $s = 2, \dots, 6$. It checks four-point orbits, order preservation, divisibility by four and the exact integer multiplicity distributions underlying the character symmetries. The new verification logs and runtime metadata are supplied with the ancillary files; these finite checks supplement the paper proof of Proposition 5.18. First, M_s has degree $n = 2s$, and

$$X^{2^n} \equiv X \pmod{M_s}, \qquad \gcd(M_s, X^{2^{n/r}} - X) = 1 \quad (r \mid n \text{ prime}),$$

which certify irreducibility. Second, $b^A = w^A = 1$, $b^{A/p} \neq 1$ for every prime $p \mid A$, $c = 1+b+w \neq 0$, and $\mathrm{Tr}_{E/\mathbb{F}_2}(b/c^2) = 1$. Third, in the quadratic field one checks

$$a^d = (1+a)^d = 1, \quad a^B = b, \quad (1+a)^B = w, \quad a^{d/p} \neq 1 \quad (p \mid d \text{ prime}).$$

Primality is certified without probable-prime assumptions: the base case is 2; for an integer $p > 2$ one supplies a complete recursively certified factorization of $p-1$ and an integer v with $v^{p-1} \equiv 1 \pmod{p}$ and $\gcd(v^{(p-1)/r} - 1, p) = 1$ for every prime $r \mid p-1$. For any prime divisor ℓ of p, these conditions force the order of v modulo ℓ to be $p-1$, hence $\ell \geq p$ and p is prime. The independent verifier uses different field multiplication and powering implementations from the certificate finder; it requires only the Python standard library, run without the `-O` flag.

The small-prime list used in the analytic constant is

$$5, 13, 17, 29, 37, 41, 53, 61, 73, 89, 97, 101,$$
$$109, 113, 137, 149, 157, 173, 181, 193, 197, 229, 233, 241.$$

Their product is 6045408114863853991373345088496297320256672385; the integer comparison is $256^{24} < 32^8$ times this product. The verifier checks this comparison as well.

As a separate small-field audit, multiplicative-subgroup enumeration and polynomial gcds with the Möbius sieve agree on the following counts. Here N_d^{prim} has the meaning in Theorem 5.19.

s	d	T_d	$\lvert\mathcal{R}_d\rvert$	N_d^{prim}	D_d
2	85	24	24	16	8
3	585	86	72	48	24
4	4369	240	240	224	208
5	33825	1214	960	520	240
6	266305	4104	3960	2928	1992

The data also verify the genus-one Hasse bounds on every primitive norm fiber in this small range. They confirm that the diagonal $b = w$ gives no realizing slope for $s = 3, 5$. These supplementary counts do not replace the general proof. The present all-dimensional theorem uses the classical Weil bound and a finite exact certificate, and is outside the Lean formalization.

Additional double-order certificates. The file `inner_double_order_certificates.json` uses the same M_s, b as the preceding table and replaces w by b^{e_s}, with the following exponents. The root $a = Z$ of $Z^2 + (1 + b + b^{e_s})Z + b$ and its translate $1 + a$ both have exact order d, for every $2 \leq s \leq 32$.

s	e_s	s	e_s
2	1	18	1
3	4	19	5
4	1	20	1
5	4	21	5
6	3	22	1
7	1	23	8
8	4	24	4
9	26	25	1
10	1	26	3
11	7	27	1
12	1	28	1
13	1	29	1
14	8	30	1
15	4	31	1
16	4	32	3
17	7		

The independent `verify_inner_mixed.py` checks irreducibility, the trace, both norms and both nonidentity residues $a^{d/p}, (1+a)^{d/p}$ for every prime $p \mid d$. It uses the complete factorizations and recursive Lucas certificates of `inner_critical_certificates.json`; all new residues and all norm-fiber counts for $s = 2, \dots, 6$ are recorded in `inner_mixed_verification.json`. It also checks the corrected mixed Lang functions and trace multipliers at the 31 certificate points. These computations establish precisely the stated finite range. Together with Proposition 5.25 for $s \geq 33$, they complete the all-dimensional double full-order theorem. The new verifier `verify_inner_double_all.py` independently rechecks all 31 double-order certificates, including their recursive Lucas primality proofs, without calling a certificate finder.

Integer bounds for double full-order existence

The bound in Proposition 5.25 is positive when $\sqrt{t} \geq 4W(d)$. We cover $33 \leq s \leq 113$ by exact divisor upper bounds, and $s \geq 114$ by a uniform inequality for odd integers. No large residual factor is assumed to be prime.

For the uniform bound let P be the product of all 53 odd primes less than 256, namely

$$3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83,$$
$$89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151, 157, 163, 167, 173,$$
$$179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233, 239, 241, 251.$$

The exact integer comparison $256^{53} < 4096^8 P$ gives, for every odd n,

$$\frac{W(n)^8}{n} = \prod_{p^e \| n} \frac{256}{p^e} \leq \prod_{3 \leq p < 256} \frac{256}{p} = \frac{256^{53}}{P} < 4096^8.$$

Here the product on the right is over primes: each factor for a prime at least 256 is at most one, and adding missing smaller prime factors only increases the bound. Thus $W(n) < 4096 n^{1/8}$. For $s \geq 114$, $d < 2t^3$ and $t > 2^{113}$ prove the sufficient condition, as computed in Proposition 5.25.

For each $33 \leq s \leq 113$, divide A and B by every prime at most 4096, with its full multiplicity. Let j_A, j_B be the numbers of distinct small factors and R_A, R_B the residuals. All prime factors of each residual exceed 4096. If $4097^{k_A} \leq R_A < 4097^{k_A+1}$ and similarly for B, then

$$\omega(d) = \omega(A) + \omega(B) \leq r_s := j_A + k_A + j_B + k_B.$$

The following table lists all 81 bounds; each satisfies $s \geq 4 + 2r_s$, so $\sqrt{t} \geq 4\,2^{r_s} \geq 4W(d)$.

s	r_s	s	r_s	s	r_s
33	11	60	15	87	24
34	9	61	16	88	21
35	13	62	15	89	23
36	11	63	21	90	27
37	11	64	15	91	26
38	11	65	20	92	23
39	13	66	19	93	25
40	9	67	18	94	24
41	11	68	16	95	26
42	14	69	19	96	24
43	12	70	20	97	26
44	11	71	18	98	26
45	17	72	19	99	29
46	13	73	20	100	26
47	13	74	19	101	26
48	12	75	23	102	27
49	15	76	19	103	26
50	14	77	23	104	25
51	15	78	22	105	32
52	12	79	21	106	27
53	14	80	19	107	28
54	16	81	23	108	29
55	16	82	21	109	27
56	13	83	22	110	29
57	16	84	23	111	30
58	15	85	24	112	27
59	15	86	22	113	29

The full small factorizations, residual integers and their integer power bounds are supplied in `inner_double_omega_certificates.json`. The independent verifier `verify_inner_double_all.py` reconstructs every small prime by deterministic trial division, checks the exact products and all 81 inequalities, and checks $256^{53} < 4096^8 P$. Its results, log and runtime information are `inner_double_all_verification.json`, `inner_double_all_run.log` and `double_all_verification_environment.json`. It also exhaustively checks the trace-one subfamily for $2 \leq s \leq 6$: its counts are respectively $8, 0, 16, 0, 24$. In particular the all-dimensional claim is for D_d, not for Δ_t. The disjoint ranges $2 \leq s \leq 32$, $33 \leq s \leq 113$, and $s \geq 114$ cover every integer $s \geq 2$.

Bases for the coordinate presentation

For the coordinate presentation $T(x) = (2^{2k} - 1) \oplus \mathrm{coords}_G(x)$ on $2k$ bits the following bases give the stated group order $\langle T, D \rangle$ and K-cycle type:

k	basis G	$\langle T, D \rangle$	K type	factor
1	$(1, 3)$	$12 = 4 \cdot 3$	$1 + 1 \cdot 3$	Φ_3
2	$(1, 2, 5, 13)$	$240 = 16 \cdot 15$	$1 + 3 \cdot 5$	Φ_5
3	$(63, 33, 18, 49, 1, 21)$	$4032 = 64 \cdot 63$	$1 + 7 \cdot 9$	Φ_9
4	$(134, 136, 241, 71, 34, 65, 3, 143)$	$65280 = 256 \cdot 255$	$1 + 15 \cdot 17$	a degree-8 factor of Φ_{17}

All four rows were verified by independent computation. The 720-case count was reproduced by enumerating all orderings of the six displayed vectors, solving the corresponding $\mathbb{F}_2$ linear system

for coords$_G$, and recording the cycle types of T, D, K whenever the two inverses exist.

Explicit τ-collapse data

With $\mathbb{F}_{2^{2k}} = \mathbb{F}_2[t]/(f)$ and elements written as polynomials in t, the following rows realize the order conditions of Theorem 6.3:

k	f	γ	$\tau = \gamma \oplus \gamma^{-1}$	$\alpha = (1 \oplus \gamma)^{-1}$
1	$t^2 + t + 1$	t	1	t
2	$t^4 + t + 1$	t^3	$t^2 + t + 1$	t
3	$t^6 + t + 1$	$t^2 + t$	$t^4 + t^3 + 1$	$t^5 + t^3 + t^2$
4	$t^8 + t^4 + t^3 + t^2 + 1$	$t^3 + t^2 + t + 1$	$t^7 + t^4 + t^3 + 1$	$t^6 + t^4 + t^3 + t^2 + 1$

In each row $\mathrm{ord}(\gamma) = \mathrm{ord}(1 \oplus \alpha^{-1}) = 2^k + 1$, τ is primitive in $\mathbb{F}_{2^k}$, and $\mathrm{ord}(\alpha) = \mathrm{ord}(1 \oplus \alpha) = 2^{2k} - 1$ (all verified). The defining polynomial f is a field-representation choice and need not coincide with the characteristic polynomial p_T of (1); for instance, the $k = 3$ row uses $f = t^6 + t + 1$, while $p_T(t) = t^6 + t^5 + t^3 + t^2 + 1$.

B Self-contained proof for the even-characteristic case

Proof of Theorem 6.2. The statement is classical (see [21] and [22, Theorem 1, p. 1]; it also follows from the equivalent square-order formulation in [20, Theorem 1.5, p. 170]); we give a self-contained proof in the even-characteristic case used above. Cohen and Presern [23] also give a self-contained treatment of prescribed traces. This appendix reconstructs the classical trace-one argument and makes its finite verification explicit; no priority is claimed for the existence result or the character-sum method. Put $q = 2^k$, $L = \mathbb{F}_{q^2}$, $K_0 = \mathbb{F}_q$, and $M = q^2 - 1 = (q-1)(q+1)$. Let

$$E_1 = \{x \in L : x^q + x = 1\}.$$

The trace map $L \to K_0$ is surjective and K_0-linear, so $|E_1| = q$. If $x \in E_1$, then $x \notin K_0$ and its minimal polynomial over K_0 is $X^2 + X + xx^q$. Hence it is enough to prove that E_1 contains an element of order M.

Let ψ be the nontrivial additive character of $\mathbb{F}_2$, and put $\psi_L(x) = \psi(\mathrm{Tr}_{L/\mathbb{F}_2}(x))$ and $\psi_{K_0}(a) = \psi(\mathrm{Tr}_{K_0/\mathbb{F}_2}(a))$. For a multiplicative character χ of L^*, extend χ to 0 by $\chi(0) = 0$ and define

$$g_L(\chi) = \sum_{x \in L^*} \chi(x)\psi_L(x), \qquad g_{K_0}(\chi) = \sum_{a \in K_0^*} \chi(a)\psi_{K_0}(a), \qquad S(\chi) = \sum_{x \in E_1} \chi(x).$$

If χ is nontrivial on L^*, then $|g_L(\chi)| = q$; if $\chi|_{K_0^*}$ is nontrivial, then $|g_{K_0}(\chi)| = \sqrt{q}$. Indeed, the usual squared-modulus calculation gives

$$|g_L(\chi)|^2 = \sum_{t \in L^*} \chi(t)^{-1} \sum_{a \in L^*} \psi_L(a(1-t)) = q^2,$$

and the same argument in K_0 gives q.

We first compute $S(\chi)$. For $\alpha \in K_0$ set $E_\alpha = \{x \in L : x^q + x = \alpha\}$; these sets partition L, and $E_\alpha = \alpha E_1$ for $\alpha \neq 0$. If $\chi^{q+1} \neq 1$, then the restriction of χ to K_0^* is nontrivial. Since $E_0 = K_0$, its contribution is $\sum_{x \in E_0} \chi(x) = 0$; hence

$$g_L(\chi) = \sum_{\alpha \in K_0} \psi_{K_0}(\alpha) \sum_{x \in E_\alpha} \chi(x) = S(\chi) g_{K_0}(\chi),$$

so $|S(\chi)| = q/\sqrt{q} = \sqrt{q}$.

Suppose now that $\chi^{q+1} = 1$ and $\chi \neq 1$. Then $\chi|_{K_0^*} = 1$. Let $H = \{h \in L^* : h^{q+1} = 1\}$; since $\gcd(q-1, q+1) = 1$, the group L^* is the direct product $K_0^* \times H$. Write $x = ch$ with $c \in K_0^*$ and $h \in H$. The condition $x^q + x = 1$ becomes $c(h + h^{-1}) = 1$; thus $h \neq 1$ and $c = (h + h^{-1})^{-1}$. This gives a bijection between $H \setminus \{1\}$ and E_1, and therefore

$$S(\chi) = \sum_{h \in H,\ h \neq 1} \chi(h) = \sum_{h \in H} \chi(h) - 1 = -1,$$

because $\chi|_H$ is nontrivial.

We shall use the primitive-element indicator

$$\mathbf{1}_{\mathrm{ord}(x)=M} = \frac{\varphi(M)}{M} \sum_{\delta \mid M} \frac{\mu(\delta)}{\varphi(\delta)} \sum_{\mathrm{ord}(\chi)=\delta} \chi(x). \tag{41}$$

To see this, fix a generator g of L^* and write $x = g^a$. The left side is the indicator of $\gcd(a, M) = 1$. For a character with $\chi(g) = e^{2\pi i j/M}$ and order $\delta = M/\gcd(M, j)$, its unnormalized Fourier coefficient is the Ramanujan sum $c_M(j) = \mu(\delta)\varphi(M)/\varphi(\delta)$; Fourier inversion gives (41). (The displayed identity for $c_M(j)$ follows by Möbius inversion.)

Assume for contradiction that E_1 has no element of order M. Summing (41) over E_1 gives

$$0 = \frac{\varphi(M)}{M} \sum_{\delta \mid M} \frac{\mu(\delta)}{\varphi(\delta)} S_\delta, \qquad S_\delta = \sum_{\mathrm{ord}(\chi)=\delta} S(\chi).$$

Let $s = \omega(M)$ and $s' = \omega(q+1)$ be the numbers of distinct prime divisors. The $\delta = 1$ term is q. If $\delta \mid q+1$ and $\delta > 1$, then each character of order δ satisfies $\chi^{q+1} = 1$, so the corresponding contribution is $-\mu(\delta)$; summing over $\delta \mid q+1$, $\delta > 1$, gives 1. For the remaining squarefree divisors $\delta \nmid q+1$, the character sums are bounded by $\sqrt{q}$, and their number is at most $2^s - 2^{s'}$. Consequently,

$$q + 1 \leq (2^s - 2^{s'})\sqrt{q}. \tag{B.1}$$

Since $q + 1 > \sqrt{q}$, this implies

$$2^s > 2^{k/2}, \qquad \text{so} \qquad s > k/2.$$

In particular $k < 2s$. Let $p_1 < p_2 < \cdots$ be the primes and $P_s = p_1 \cdots p_s$. Since $M = 2^{2k} - 1$ has exactly s distinct prime divisors, $M \geq P_s$. We have

$$P_{16} = 32589158477190044730 > 2^{64}.$$

For $n \geq 16$, the ratio

$$\frac{P_{n+1}/2^{4(n+1)}}{P_n/2^{4n}} = \frac{p_{n+1}}{16}$$

is greater than 1, because $p_{n+1} \geq 59$. Hence $P_s > 2^{4s}$ for every $s \geq 16$. But $k < 2s$ gives

$$M = 2^{2k} - 1 < 2^{2k} < 2^{4s},$$

so $s \geq 16$ is impossible. Thus $s \leq 15$ and $k < 2s \leq 30$, i.e. $k \leq 29$.

It remains to cover this finite range. Each row below specifies an irreducible degree-k polynomial m_k over $\mathbb{F}_2$ and an element $a_k \in K_0$ with $\mathrm{Tr}_{K_0/\mathbb{F}_2}(a_k) = 1$, where $K_0 = \mathbb{F}_2[x]/(m_k)$. The displayed

hexadecimal integers encode their binary coefficient vectors. For each row let t be a root of $X^2 + X + a_k$ in $K_0[t]/(X^2 + X + a_k)$. The deterministic verifier checks that m_k is irreducible and that $\mathrm{Tr}_{K_0/\mathbb{F}_2}(a_k) = 1$. The trace condition implies that $X^2 + X + a_k$ has no root in K_0 and is irreducible there; the quotient is therefore a quadratic field extension. The verifier then checks that $t^{2^k} = t + 1$, and that $t^M = 1$ while $t^{M/p} \neq 1$ for every prime $p \mid M$. The last two checks show that $\mathrm{ord}(t) = M$; the third shows that $\mathrm{Tr}_{L/K_0}(t) = 1$. A verifier for the table, `verify_k1_29.py`, is included as an ancillary file. These checks are performed row by row using polynomial arithmetic in characteristic two. Its SHA-256 hash is `042CE7F74EDFF13ADE5C6285E42CB3D15491FC25C65B99BC670D256047EE8D9F`.

k	m_k	a_k
1	0x3	0x1
2	0x7	0x3
3	0xb	0x5
4	0x13	0xd
5	0x25	0x13
6	0x43	0x34
7	0x83	0x47
8	0x11b	0xc8
9	0x203	0x2b
10	0x409	0xcb
11	0x805	0x525
12	0x1009	0x39e
13	0x201b	0x1ec4
14	0x4021	0x3717
15	0x8003	0x62bf
16	0x1002b	0x1e6e
17	0x20009	0x1155b
18	0x40009	0x2ee84
19	0x80027	0xb9af
20	0x100009	0x6aee2
21	0x200005	0x15750b
22	0x400003	0x24a150
23	0x800021	0x6c92f9
24	0x100001b	0x389ec4
25	0x2000009	0x6801ad
26	0x400001b	0x2707d2e
27	0x8000027	0x6e9ab1e
28	0x10000003	0xfac7cd2
29	0x20000005	0xc69b55e

The table therefore supplies the required element for every $k \leq 29$. Combining this with the contradiction for $k \geq 30$ proves the theorem. □